# How Long Does a Gold Rush Last?

Evidence from Africa's Gold Price Cycle

Albert Duodu*

November 2026

**Abstract**

Artisanal gold mining spread across Africa as the gold price rose until 2012. How long does such a gold rush last? I exploit the interaction between geology suitable for artisanal gold mining and the rise, fall and second rise of the world gold price, using household surveys from 33 African countries and satellite data. In the eight countries surveyed on both sides of the 2012 peak, men in a fully gold-suitable place leave farming as the price rises, by 16 percentage points at the peak, and return to it when the price falls, recovering 9.5 points. When the price climbed past its earlier peak after 2019, men did not leave farming again. Forest loss rises with the price, stays high after the peak and rises further after 2019. Teenagers in gold-suitable places attend school less during the boom, but completed schooling shows no detectable loss, and household wealth keeps most of its gain.

**JEL codes:** I25, J24, O13, O17, Q23, Q32.

**Keywords:** artisanal mining, gold, resource booms, structural change, deforestation, Africa.

*Aalto University, Department of Economics. albert.duodu@aalto.fi.

# 1 Introduction

Between 2000 and 2012, the world price of gold rose from $279 to $1,670 an ounce, and across Africa people moved into artisanal mining, a labour-intensive and largely informal activity on which 130 to 270 million people depend worldwide (Girard et al., 2025). Then the price fell. By 2015 it was 30 percent below its peak, and it stayed near that level until 2018, before rising again to a new record by 2024. What happened to the people the gold rush had drawn in, and to the places it had changed?

Recent work shows what the boom did to gold-suitable places: it cleared forest and raised household wealth (Girard et al., 2025), lowered school attendance among adolescents (Ghazarian and Khan, 2025), raised women's employment in mining and trade and reduced violence by their partners (Byambasuren and Nguyen, 2026), and created new small towns (Girard and Pignède, 2026). A full price cycle shows how long these changes last. Artisanal mining needs little more than labour and tools, so people can leave it as cheaply as they entered, and work should follow the price in both directions. Other changes need not reverse. Cleared forest does not regrow when miners leave, a teenager who leaves school for the pits may not go back when the pits close, and assets bought in the boom outlast it. Who bears each change also differs, because men, women and children enter the gold economy in different ways.

Two features of the setting make the question hard to answer. Most African countries keep no record of where artisanal mining happens, and few household surveys code mining separately. And through most of the years that recent papers study, the gold price rose steadily: its correlation with the calendar year over 1990 to 2018 is 0.83, so a differential trend in gold-suitable places would look like a response to the price. The fall after 2012 and the rise after 2019 separate the two, because a trend continues in one direction while a response to the price turns with it.

In this paper, I follow work, land use, schooling and household wealth in gold-suitable places through the full price cycle. Following Girard et al. (2025), I measure exposure by the share of each half-degree cell's land on geology suitable for artisanal gold mining, and I compare cells within the same country and year as the lagged world price moves. Geology predates the boom and no African mining area sets the world price, so the variation is plausibly exogenous. I compare the peak of 2012–14 with the years of low prices that followed, in the eight countries surveyed on both sides of the peak, and the years of low prices with the second rise of 2019–25.

I build the household data from 166 Demographic and Health Surveys fielded in 33 countries between 1986 and 2025. In the 132 that record cluster coordinates, 1.6 million women, 0.7 million men and 1.6 million households are linked to their half-degree cell. The data include the men's own interviews, which record each man's occupation, the surveys fielded after 2018, household rosters with school attendance and completed schooling, the survey's wealth index and the birth histories of mothers. Forest loss,

night lights and the footprint of mines come from satellite data.

First, men's work leaves farming with the price and returns to it. Across all countries, farming in gold-suitable places falls as the price rises, by about 4 percentage points over the rise to the peak in a fully gold-suitable cell, and men and women leave it in equal measure. In the eight countries surveyed on both sides of the peak, men's farming is 16.1 points lower at the peak than before the boom, on a mean of 41, and it recovers 9.5 points when the price falls, mostly as farming for part of the year; their manual work, where most artisanal miners are recorded, rises by 5.6 points and gives up 4.9. These are changes in a fully gold-suitable cell; in the average cell with suitable land, where the share is 0.43, they are about two-fifths as large. The farming cycle holds in rural clusters alone and in cells surveyed in every window. It also holds with controls for urban residence, for industrial mines of other minerals at their own world prices, for armed conflict and for rainfall, and with standard errors that allow correlation across cells up to 500 kilometres apart. Women move into sales, which rise by 5.5 points and give up 4.0, and the share of women in any work falls by 8 points at the peak, on a mean of 57.

Second, the second rise did not draw men out of farming again. In the ten countries surveyed both after the peak and in 2019–25, the lagged price at the interview dates was 16 percent higher than in the peak surveys. Yet between the two windows, men's farming in a fully gold-suitable cell rose by 2.0 points, and the estimate rules out a fall of more than 2.5 points, against the fall of 16 points at the first peak. Men's manual work did not move, and women's sales did not rise again.

Third, forest clearing does not slow when the price falls. Forest loss rises with the price, and at the peak it is 0.20 log points higher in a fully gold-suitable cell than in 2001–05. It does not fall when the price drops, and in the second rise it grows by a further 0.06 log points. Year by year, forest loss follows the price beyond a linear trend in gold-suitable land: with a linear trend for these cells, a unit rise in the price index still raises loss by 0.17 log points. The further clearing lies away from the mines, in cells with no record of gold or artisanal mining and with no mine on a satellite map of mine footprints (Tang and Werner, 2023), and on a global map of the drivers of forest loss most of the extra clearing lies in pixels attributed to farming, shifting cultivation and logging.

Fourth, the boom leaves a longer trace on places than on work. Compared with children aged 6 to 11 in the same cell and survey, whom school closures or mercury would affect as well, teenagers aged 16 to 20 attend school 4.9 points less at the peak, on an attendance rate of 47 percent. The gap narrows after the peak, to about half its size, and stays there through the second rise, though neither later gap is precisely estimated. Completed schooling shows no detectable loss: among adults aged 21 to 45, years of schooling do not fall with the price their cohort faced at ages 12 to 20. The estimate excludes the loss of about 0.4 years that the attendance gaps at the peak would

imply had they lasted through a cohort's teenage years. The wealth index rises by 0.25 standard deviations at the peak and keeps three-fifths of the gain; in rural households the rise is 0.17, and the urban share of households rises at the peak, so part of the gain may come with a shift toward towns. Night lights keep rising after the peak, although from 2006 on a trend for gold-suitable cells fits that rise about as well as the price does, and child mortality does not change on average; in one of several splits by group, it rises in towns that predate the boom.

Before turning to the data, I note four limits of this evidence. First, the cycle in work rests on the eight countries surveyed on both sides of the peak, and in two of them, Chad and Rwanda, the peak and the years after it come from one survey fielded across 2014 and 2015, so six countries carry most of the comparison, and governments and donors, not the gold price, set the years in which surveys were fielded. Ghana and Burkina Faso, two of the largest artisanal gold producers, and Côte d'Ivoire enter neither the comparison across the peak nor that of the second rise. With each country as one cluster, intervals widen, because survey rounds within a country differ by more than sampling error; men's fall from farming at the peak remains statistically significant, but the recovery is less precise (Section 6). Dropping each country in turn leaves the recovery of men's farming between 7.9 and 10.6 points, and the five countries that enter both comparisons show the same cycle. Second, part of the cycle is common to whole regions or follows the timing of fieldwork. With region-by-year effects or with effects for the month of interview, men's farming still falls and recovers, but the fall in manual work and sales after the peak is small. When the suitability map is shifted at random across the continent, cycles in farming as large as the actual one appear in 5 to 7 percent of the shifted maps, and cycles in manual work and sales in 13 to 32 percent. Third, forces other than the price changed over the same years in mining areas. The depletion of shallow deposits, crackdowns on illegal mining, the cost of mercury and fuel, and the food-price shock of 2021 and 2022 could each lower mining after the peak or mute the response to the second rise, so the return to farming and the flat second rise need not reflect the price alone. The flat second rise also appears in cells without a recorded history of gold mining, which is hard to reconcile with depletion alone, but I have no data on enforcement or on mining equipment across the ten countries. Fourth, the surveys are repeated cross-sections, so the persistence in schooling and wealth describes places, and the wealth index ranks each household within its survey. Without a window before the boom long enough to fit a trend, the persistence of schooling, wealth and night lights cannot be told apart from a trend for gold-suitable places that began with the boom. Still, a full price cycle gives a test that a rising price cannot, of whether each change follows the price back down, and work, land and schooling answer it differently.

These results add to work on local resource booms. Large mines raise local incomes through their demand for inputs (Aragón and Rud, 2013) and move women from farm-

ing into services (Kotsadam and Tolonen, 2016), and Huang et al. (2026) find that higher mineral prices shift work from agriculture to services around mines worldwide. Much of the evidence on what happens when a boom ends comes from the United States, where wages and employment in oil and gas counties rise in booms and fall in busts (Allcott and Keniston, 2018), and incomes in the boomtowns of the American West fell below their earlier path once the boom ended (Jacobsen and Parker, 2016); Black et al. (2005) follow local labour markets through the coal boom and bust of the 1970s and 1980s. In Britain, the collapse of coal mining left a lasting fall in men's employment (Aragón et al., 2018), while the gains from the fracking boom spread well beyond the producing counties (Feyrer et al., 2017) and oil specialisation in the American South raised population and income over decades (Michaels, 2011). In Africa, Kotsadam and Tolonen (2016) find that the service and skilled jobs that open with large mines contract when the mines close. Large mines also raise local wealth while worsening some health outcomes (von der Goltz and Barnwal, 2019), lower infant mortality nearby (Benshaul-Tolonen, 2019) and, through pollution, lower agricultural productivity around them (Aragón and Rud, 2016). I show that men's work in Africa's gold-suitable places leaves farming with the price and returns to it when the price falls, while forest clearing does not fall back, and that a later boom raised clearing again without drawing men out of farming.

The schooling results join work on how local labour demand shapes education. Higher returns to work can draw children out of school, as coffee booms did in Brazil (Kruger, 2007) and good rainfall years did in India (Shah and Steinberg, 2017), although higher household income can work the other way, as rising rice prices did in Vietnam (Edmonds and Pavcnik, 2005). Export factories in Mexico raised school dropout (Atkin, 2016), the fracking boom in the United States reduced schooling among young men (Cascio and Narayan, 2022), and the housing boom in the United States lowered college attendance in ways that persisted after the bust (Charles et al., 2018). Ghazarian and Khan (2025) show that the rise in the gold price lowered school attendance among adolescents near gold-suitable land, and completed schooling among young adults who faced a high price as children. I find that lower attendance in gold-suitable places outlasts the shift in work, since the attendance gap only narrows, imprecisely, when the price falls, while the completed schooling of the cohorts who were teenagers during the boom shows no detectable loss.

The closest papers use the same geology and the world gold price. Girard et al. (2025) show that the boom raised deforestation and household wealth, with no significant change in work in agriculture, Byambasuren and Nguyen (2026) that it raised women's employment and reduced violence by their partners, and Girard and Pignède (2026) that it created small towns without structural change; Byambasuren and Nguyen (2026) also relate violence to the gold price up to ten years earlier and find that the fall in intimate partner violence lasts two years. In Ghana, Duodu and Danquah (2026) com-

bine gold-bearing Birimian geology with the gold price and find that fully gold-suitable districts host 19 percent fewer formal firms. Bazillier and Girard (2020) compare the two kinds of mining in Burkina Faso and find that household consumption rises with the gold price near artisanal mines but not near industrial ones. Each estimates one response to the price, mostly over years in which it rose. Girard et al. (2025) use surveys up to 2018, which include the first years after the peak, but impose one slope on them, for women and the occupations of their partners; Girard and Pignède (2026) let the response differ by period but note that the price path they study does not allow them to assess a collapse. I follow the same places, with men's own occupations, through the fall that followed the peak and the rise that followed the fall, and I find that work, land, schooling, wealth and health respond to the cycle on different clocks.

Section 2 describes artisanal mining and the price cycle. Sections 3 and 4 describe the data and the empirical strategy. Section 5 reports the results and Section 6 their robustness. Section 7 concludes.

## 2 Artisanal Gold and the Price Cycle

### 2.1 Artisanal Gold Mining in Africa

"Artisanal and small-scale mining refers to formal or informal mining operations with predominantly simplified forms of exploration, extraction, processing, and transportation" (OECD, cited in Girard et al., 2025). It employs an estimated 44.75 million people worldwide (World Bank, 2020), and in Africa its main mineral is gold (Girard et al., 2025). The tools are simple. Miners dig pits and shafts, or wash river gravel, with picks, shovels, pans and sluices, and they often separate the gold with mercury. Most miners work close to home, many in the dry season between harvests (Bazillier and Girard, 2020), and many without a licence.

Men do most of the digging. Women are a large share of the workforce, estimated at 40 to 50 percent of artisanal miners in Africa, and many work in processing, including amalgamation with mercury (Hinton et al., 2003). Adolescents work on sites as well, and school attendance near gold-suitable land falls when the gold price rises (Ghazarian and Khan, 2025). Few household surveys record this work under its own name. In the occupation codes of the Demographic and Health Surveys, most diggers appear as manual workers.

Industrial gold mines differ in scale and in their links to the local economy. In Burkina Faso in 2014, an estimated 640,800 people worked as gold diggers, while the industrial mining sector reported 6,464 employees (Bazillier and Girard, 2020). Industrial mines are capital-intensive projects licensed by the state, run by international companies with skilled employees on formal contracts, and they import most of their equipment (Bazillier and Girard, 2020); their local effects run mainly through their purchases of

local inputs (Aragón and Rud, 2013). The two kinds of mining often sit on the same geology: of the 139 half-degree cells that ever host an industrial gold mine, 133 contain land suitable for artisanal mining.

Artisanal mining has also changed its technology. In Ghana, tens of thousands of Chinese miners arrived from the late 2000s (Hilson et al., 2014) and brought excavators and large wash plants to artisanal sites, which turned over far larger areas of land than digging by hand (Crawford and Botchwey, 2017). Where machines do the digging, a rise in the price can raise output and the land cleared without a matching rise in employment. The machines also drew the state in. In 2017 Ghana suspended small-scale mining, and soldiers sent to enforce the ban burned excavators at illegal sites (Hilson, 2017). Illegal mining also responds to the incentives of local governments: in Colombia it rose where a reform cut the royalties that municipalities received (Saavedra and Romero, 2021).

### 2.2 The Gold Price Cycle

After two decades of decline, the world price of gold stood at $279 an ounce in 2000. It rose from 2001, nearly tripled between 2005 and 2010 and peaked at an annual average of $1,670 in 2012. Then it fell. By 2015 it was 30 percent lower, at $1,161, a fall of 22 percent in real terms, and it stayed close to that level until 2018. The second rise began in 2019: the price reached $1,770 in 2020 and $2,388 in 2024, 43 percent above its 2012 level, or 37 percent in real terms (World Bank Pink Sheet).

Figure 1 shows the lagged price index that I use, with 2013 equal to one, and the five windows that organise the analysis, around the reference period of 2000–05. Averaged over calendar years, the index is 0.23 in 2000–05, the reference period; 1.10 at the peak, 2012–14; 0.87 after the peak, 2015–18; and 1.25 in the second rise, 2019–25. The rise between 2006 and 2011 and the years before 2000 complete the path.

Household surveys cover the cycle unevenly (Figure 1, Panel B). Each country fields a survey only every four to six years, so only eight countries were surveyed both at the peak and after it: Benin, Chad, Guinea, Mali, Nigeria, Rwanda, Senegal and Zambia. Ten were surveyed both after the peak and in 2019–25, five of them among the eight. The first group carries the comparison of the peak with the years after it; the second carries the comparison of those years with the second rise.

The setting suggests how each outcome should move over the cycle. Men's work should follow the price in both directions, because entry and exit cost little. Land cleared for pits does not regrow when miners leave, and assets bought during the boom outlast it. Where excavators replace diggers, a later rise in the price can raise output and the land cleared without drawing workers back. For teenagers, the boom raises the return to work against school; whether they return to school when the price falls depends on how hard it is to resume after leaving. Section 5 takes these outcomes in this order.

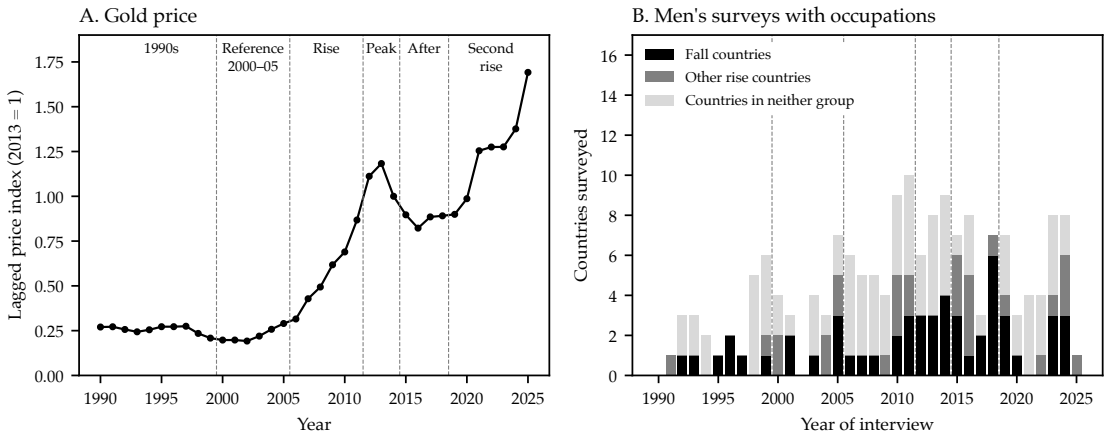



Figure 1—The gold price and the timing of the surveys

*Notes*: Panel A plots the gold price index lagged by one year (nominal, 2013 equal to one), with the windows used in Section 4. Panel B counts countries with a men's survey recording occupations in each interview year. Fall countries are the eight surveyed both at the peak and after it; other rise countries are those surveyed after the peak and in 2019–25 that are not fall countries; the remaining countries are in neither group. Sources: World Bank Pink Sheet; Demographic and Health Surveys.

# 3 Data

I combine a map of geology suitable for artisanal gold mining, household surveys that I assemble and link to the map, and satellite data on forest loss. Appendix A describes the construction of each file.

## 3.1 Gold-Suitable Land

Most African countries keep no register of artisanal mines, and where a site opens may depend on local labour supply. Following Girard et al. (2025), I measure where artisanal gold could be mined. Their map classifies bedrock across Africa as suitable for artisanal gold mining by its age, chemistry and structure, and it marks about 18 percent of the continent (Figure 2, Panel A). For each of the 10,686 half-degree PRIO-GRID cells of the continent (Tollefsen et al., 2012), I compute the share of its area on that bedrock. The share is positive in 4,514 cells, where it averages 0.43 (Panel B). Exposure is this share times the world gold price index lagged by one year: it is zero in a cell without suitable land in every year and rises with the price in a cell with it. The map captures where artisanal mining takes place. Other records of gold or artisanal gold mining, from exploration surveys, administrative and survey records of sites, mercury emissions and sites detected in satellite images, exist in 44 percent of cells with suitable land and in 8 percent of the others (Panel C). On the satellite map of mine footprints of Tang and Werner (2023), a fully gold-suitable cell with no industrial gold mine is 5.4 percentage points more likely to contain a mine than a cell without suitable land, against an average of 2.7 percent (Appendix Table B8, Panel G), and the artisanal mercury emissions that Steenhuisen and Wilson (2019) allocate across space are higher there as well. Because

industrial gold mines sit on the same geology (Section 2), Section 6 adds an industrial-mine indicator times the same price index to the main cycle estimates.

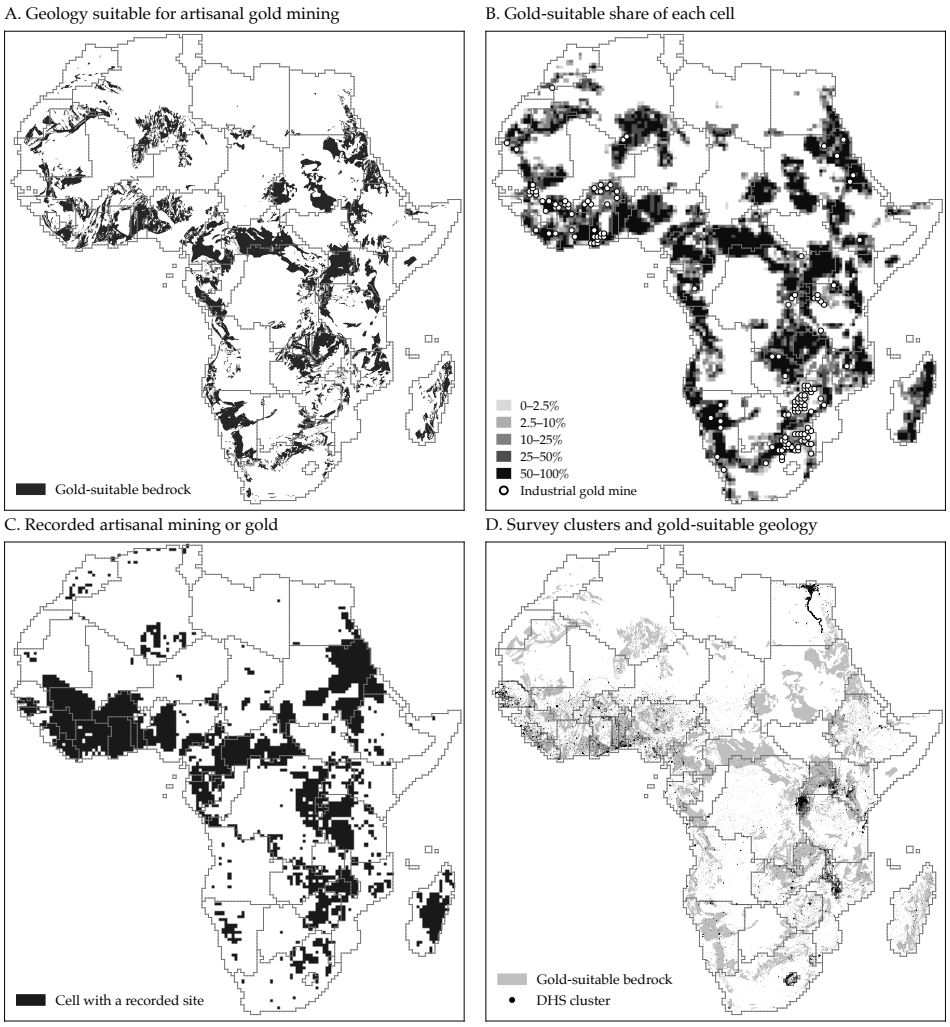



Figure 2—Where artisanal gold can be mined

*Notes*: Panel A maps the bedrock that Girard et al. (2025) classify as suitable for artisanal gold mining. Panel B shades each half-degree cell by the share of its land on that bedrock, which is the exposure measure; white cells have none, and circles mark cells that ever host an industrial gold mine. Panel C marks cells with any other record of gold or artisanal gold mining in the data of Girard et al. (2025): exploration records, gold deposits, administrative and survey records of artisanal sites, mercury emissions and artisanal sites detected in satellite images. Such records exist in 44 percent of cells with gold-suitable land and in 8 percent of other cells. Panel D overlays the 65,746 survey clusters of the Demographic and Health Surveys used in the paper on the bedrock of Panel A; cluster coordinates are displaced by up to 5 kilometres in rural areas and, for a few, up to 10. Country borders follow the PRIO-GRID cells (Tollefsen et al., 2012); industrial gold mines and the records of Panel C are from the data of Girard et al. (2025).

## 3.2 Household Surveys

The household data come from the Demographic and Health Surveys (ICF, 2026), which I assemble from the raw files of every standard survey in 33 African countries: 166 surveys fielded between 1986 and 2025. The surveys form repeated cross-sections, so I

follow places over time. The 132 surveys that record the coordinates of their sampling clusters locate 65,746 clusters, and I assign each cluster to its half-degree cell (Figure 2, Panel D); 99 percent of interviewed people in these surveys are linked to a cell. To protect respondents, DHS displaces urban clusters by up to 2 kilometres and rural clusters by up to 5 kilometres, and one percent of rural clusters by up to 10 (Burgert et al., 2013). Displacement adds noise to exposure without shifting it toward gold-suitable land, and 26 percent of clusters lie close enough to a cell edge that displacement could have moved them across it. I flag them and drop them in Section 6.

Working from the raw files gives three things that the survey file of Girard et al. (2025) lacks: the men's interviews, in which each man reports his own occupation; the surveys fielded after 2018; and the household rosters, which record school attendance for every member.[1]

### 3.3 Outcomes

Work comes from the occupation each respondent reports, coded into the survey's standard groups. Each work and attendance outcome is an indicator equal to one if the person works in the sector or attends school, which I multiply by 100, so that coefficients are in percentage points. Manual work is skilled or unskilled manual work, where most artisanal miners are recorded, and agriculture includes farming on one's own account and as an employee. In the surveys whose detailed occupation codes name mining without pooling it with other trades, I also measure mining directly. For men, the outcomes are their own occupations: 664,621 men in 119 surveys from 32 countries. For women, I use their own occupation and whether they currently work, for 1.64 million women in 130 surveys, and the measure of Girard et al. (2025), which equals one if the woman or her partner works in the sector. School attendance is attendance at any time in the current school year, recorded for every member of the household. Household wealth is the survey's index of assets and housing, which DHS standardises within each survey; I compare it within surveys only, for 1.36 million households in 101 surveys from 2003. Table 1 reports the means of each outcome.

### 3.4 Forest Loss

Forest loss comes from Global Forest Change, version 1.12, which maps tree cover in 2000 and its loss in each year from 2001 to 2024 at a resolution of 30 metres (Hansen et al., 2013). I aggregate it to cells as the area of canopy above 30 percent in 2000 and the area of that forest lost each year, and I keep the 4,626 cells with at least one square kilometre of forest in 2000. Measured loss across Africa doubles between 2012 and 2014, in part because the product detects loss better from those years. Country-by-year effects

[1] Ethiopian surveys before 2024 record dates in the Ethiopian calendar, which I convert to the Gregorian calendar. In six surveys more than 40 percent of couples have no sector recorded, and I treat their occupations as missing.

Table 1—Summary statistics

| | All | Gold-suitable cells | Other cells |
|---|---|---|---|
| *Panel A: Men, own occupation* | | | |
| Gold-suitable share of cell | 0.28 | 0.45 | 0.00 |
| | [0.33] | [0.31] | [0.00] |
| Manual work (%) | 20.0 | 19.4 | 20.9 |
| Agriculture (%) | 39.1 | 41.4 | 35.5 |
| Mining, named in occupation (%) | 0.81 | 0.93 | 0.59 |
| Age | 30.9 | 30.8 | 31.1 |
| | [12.0] | [11.9] | [12.2] |
| Men | 664,621 | 409,939 | 254,682 |
| Surveys | 119 | 115 | 112 |
| *Panel B: Women* | | | |
| Currently working (%) | 54.4 | 61.2 | 46.0 |
| Agriculture, own occupation (%) | 25.4 | 32.5 | 16.5 |
| Manual work, own occupation (%) | 7.2 | 8.0 | 6.4 |
| Manual work, woman or partner (%) | 20.7 | 19.5 | 22.3 |
| Women | 1,640,605 | 909,101 | 731,504 |
| Surveys | 130 | 126 | 123 |
| *Panel C: Children and households* | | | |
| Attending school, ages 12–15 (%) | 75.8 | 77.8 | 73.2 |
| Attending school, ages 16–20 (%) | 46.5 | 46.5 | 46.5 |
| Children aged 12–20 | 1,454,629 | 804,361 | 650,268 |
| Wealth index (within-survey z-score) | 0.00 | −0.04 | 0.06 |
| | [1.00] | [0.96] | [1.05] |
| Bottom two national wealth quintiles (%) | 41.1 | 42.4 | 39.4 |
| Households | 1,361,516 | 796,054 | 565,462 |
| Surveys with the wealth index | 101 | 97 | 94 |
| *Panel D: Cells with forest in 2000, 2001–2024* | | | |
| Tree cover in 2000 ($km^2$) | 1,272 | 1,413 | 1,105 |
| | [1,177] | [1,166] | [1,168] |
| Forest loss (% of 2000 cover a year) | 0.57 | 0.59 | 0.54 |
| | [1.34] | [1.33] | [1.34] |
| Cells | 4,626 | 2,505 | 2,121 |

*Notes*: Means, with standard deviations of continuous variables in brackets. Counts are respondents and households with the outcomes shown; Appendix Table A1 counts every interview linked to a cell. Gold-suitable cells have any land on the bedrock of Figure 2. Panels A to C are drawn from the 132 Demographic and Health Surveys (ICF, 2026) with cluster coordinates, 1986 to 2025; the count of surveys differs by panel because not every survey has a men's file or records occupations. Occupations follow the survey's standard groups: manual is skilled and unskilled manual work, agriculture includes agricultural employees. Mining is the share of men whose survey-specific occupation names mining, quarrying or gold digging without pooling it with other trades, in the surveys that code it. The woman-or-partner measure follows Girard et al. (2025). Attendance is attendance at any time in the current school year. The wealth index is the survey's asset index, standardised within each survey. Panel D covers the cells with at least 1 $km^2$ of forest (canopy above 30 percent) in 2000 in countries with more than one such cell, the sample of Figure 4; forest loss is from Global Forest Change, version 1.12 (Hansen et al., 2013).

absorb the common change, and in Section 5.3 I also let it differ with each cell's forest cover. Night lights come from the harmonised series of Li et al. (2020), which calibrates DMSP images for 1992 to 2013 and converts VIIRS images for 2014 to 2024 to the same scale; I average them over each cell.

Table 1 compares cells with and without gold-suitable land. Men in gold-suitable cells farm more (41 against 36 percent), and women there are more likely to work (61 against 46 percent), mostly in agriculture (32.5 against 16.5 percent of all women). Men name mining as their occupation in 0.9 percent of cases in gold-suitable cells and 0.6 percent elsewhere, a small share because most miners report another occupation. Attendance at ages 16 to 20 is the same in both groups, and wealth is slightly lower in gold-suitable cells. These are differences in levels, which cell fixed effects absorb; the design compares the same cells as the price moves.

## 4 Empirical Strategy

I compare cells within the same country and year that have more or less gold-suitable land, as the world price of gold moves. For person or household $i$ in cell $c$, country $k$ and year $t$, I estimate

$$Y_{ickt} = \beta \left( S_c \times P_{t-1} \right) + X'_{ickt}\gamma + \alpha_c + \lambda_{kt} + \varepsilon_{ickt}, \tag{1}$$

where $S_c$ is the share of the cell on gold-suitable bedrock and $P_{t-1}$ is the world gold price index in the year before the interview, with 2013 equal to one. The coefficient $\beta$ measures how much more the outcome moves with the price in a fully gold-suitable cell than in a cell with no suitable land. Between 2000–05 and the peak of 2012–14, the index rose by 0.87, so $0.87\beta$ is the effect of the boom on a fully gold-suitable cell.

To follow the price through its cycle, I replace $S_c \times P_{t-1}$ with the suitable share interacted with indicators for the windows of Figure 1:

$$Y_{ickt} = \sum_{w \in \mathcal{W}} \beta_w \left( S_c \times \mathbf{1}[t \in w] \right) + X'_{ickt}\gamma + \alpha_c + \lambda_{kt} + \varepsilon_{ickt}, \tag{2}$$

where $\mathcal{W}$ contains the 1990s, the rise (2006–11), the peak (2012–14), the years after the peak (2015–18) and the second rise (2019–25, or 2019–24 for forest loss and night lights), and 2000–05 is the reference period. The reference is the trough of the price, when the lagged index averaged 0.23 and artisanal gold paid least, so each coefficient measures how far a phase of the cycle moved gold-suitable places from their position before the boom; the 1990s, with fewer surveys and a price 0.03 higher, enter as a separate window. Each $\beta_w$ is the change in the outcome between the reference period and window $w$ in a fully gold-suitable cell, relative to a cell with no suitable land. The surveys with cluster coordinates start in 1990; attendance starts in 2000 and forest loss in 2001, so for these

outcomes $\mathcal{W}$ omits the 1990s. Three comparisons carry the results. The peak coefficient measures the boom. The difference between the coefficients after the peak and at the peak measures the response to the fall, estimated in the eight countries surveyed on both sides of the peak. The difference between the second rise and the years after the peak measures the response to the second rise, estimated in the ten countries surveyed in both windows. The 1990s coefficient shows whether gold-suitable cells were already moving apart from other cells before the price rose.

Cell effects $\alpha_c$ absorb every fixed difference between places, including the differences in levels in Table 1 and the direct effect of geology. Country-by-year effects $\lambda_{kt}$ absorb national shocks and policies and the effect of the gold price on each national economy, so the main effect of the price is not identified and does not appear. The price enters with a lag because survey questions on work refer to the past year and because a change in the world price takes time to reach the price paid at a site. The controls $X_{ickt}$ are age and its square for individuals and, following Girard et al. (2025), extreme temperature and the cell's suitability for its main cash crop times that crop's lagged world price. The two cell controls end in 2018, so estimates that use the surveys after 2018 omit them, and Section 6 reports the earlier estimates with and without them. Industrial gold mines sit on the same geology, in 133 of the 139 cells that host one, so $\beta$ is the response of gold-suitable places as a whole, including any response of industrial mines there. I leave industrial mines out of equation (1) because some opened during the boom, so their presence partly responds to the price; Section 6 adds mines producing by 2000, and then every industrial mine, each interacted with the price.

The same equation applies to each outcome. Its unit is the person for men's and women's own work, the age group in a cell and survey for school attendance (equation (3)), the household for wealth and the amenities, and the cell and year for forest loss and night lights. I cluster standard errors by cell, the level at which exposure varies.[2]

Identification rests on one assumption: had the gold price not changed, outcomes in more and less gold-suitable cells of the same country would have followed the same trends. No African mining area sets the world price of gold, and the geology that decides where artisanal gold can be mined predates the boom, so neither side of the interaction responds to local conditions. The assumption could still fail if gold-suitable places were on different paths for other reasons. Because the price rose steadily for a decade, a differential trend would look like a response to the boom, which is why the results follow the price down after 2012 and up again after 2019: a trend continues in one direction, while a response to the price turns with it.

Surveys sample different people in each round, and a boom may change who lives in gold-suitable cells. Men report their own occupations, so changes in household

[2]The comparisons of the peak with the years after it rest on eight countries. For these, I also report p-values from a wild cluster bootstrap with country clusters, which enumerates all $2^8$ sign patterns (Section 6).

structure do not move the men's results, and the results for teenagers compare them with younger children in the same cell and survey; Section 6 also re-estimates the cycle for respondents who have lived in the place for at least five years. Displacement of the survey coordinates places some clusters in the wrong cell, which biases the estimates toward zero, so Section 6 drops the clusters near a cell edge as well. Because the comparisons across the peak rest on eight countries, I report each result without each country in turn.

# 5 Results

## 5.1 Farming Follows the Price

In Table 2, I report the average effects of the gold price on work, from equation (1) in all countries with surveys up to 2018, for all adults and for men and women separately. As the price rises, people in gold-suitable places leave farming. Agriculture falls by 4.9 percentage points per unit of the price index (Panel A, column 4), or by about 4.3 points over the rise to the peak, 14 percent of its mean of 31, and men and women leave it in equal measure. They go to different work. Women move into sales, by 3.8 points per unit of the index on a mean of 17 (Panel C, column 5), while men's sales do not change and services fall by about 1 point per unit of the index for both, on a mean of 4 (column 6). Manual work rises for both, by about 1 point per unit of the index, but imprecisely.

These averages pool the boom with the decades before it, and they cannot show what happened when the price fell. In Figure 3, I therefore follow all adults, men and women through the windows of the price cycle in the eight countries surveyed both at the peak and after it, with the price in each window drawn behind the estimates; Appendix Table B1 reports the coefficients.

The shifts in occupation reverse when the price falls, and farming moves most. At the peak, adults in a fully gold-suitable cell farm 12.9 percentage points less than in 2000–05, and more of them work in manual jobs (2.2 points) and in sales (4.5 points). After the peak, farming recovers 5.2 points, and manual work and sales give up 2.4 and 2.8 points, although the price after the peak stayed well above its level before the boom.

The average hides two paths. Men carry the swing between farming and manual work (Panels A and B). At the peak, their farming is 16.1 points lower on a mean of 40.8, and their manual work is 5.6 points higher, 29 percent of its mean of 19.0. With the eight countries as clusters, this fall in farming is the one cycle estimate that the wild cluster bootstrap keeps below the 5 percent level; a jackknife over countries also keeps its recovery after the peak there (Section 6).[3] After the peak, farming recovers 9.5 of its 16.1 points and manual work gives up 4.9 of its 5.6, so manual work is back near its

[3]Wild cluster restricted bootstrap with country clusters, over all 256 Rademacher draws: $p = 0.02$ for men's farming at the peak and 0.12 for their manual work, and 0.16 and 0.18 for the recovery of farming and the fall of manual work after the peak.

Table 2—Artisanal gold and work: average effects

| | Currently working (1) | Manual work (2) | Extraction (3) | Agriculture (4) | Sales (5) | Services (6) |
|---|---|---|---|---|---|---|
| *Panel A: All adults* | | | | | | |
| Suitable × lagged price | −1.32 | 1.05 | 0.29 | −4.90*** | 2.98*** | −0.95* |
| | (1.42) | (0.77) | (0.37) | (1.71) | (1.00) | (0.50) |
| Mean of outcome (%) | 61.0 | 10.3 | 0.9 | 31.1 | 14.0 | 4.2 |
| Observations | 1,691,736 | 1,657,666 | 1,116,355 | 1,657,666 | 1,657,666 | 1,657,666 |
| Cells | 4,120 | 4,120 | 3,371 | 4,120 | 4,120 | 4,120 |
| Countries | 33 | 33 | 28 | 33 | 33 | 33 |
| *Panel B: Men* | | | | | | |
| Suitable × lagged price | −0.65 | 1.16 | 0.22 | −5.04*** | 0.55 | −0.61 |
| | (1.73) | (1.11) | (0.63) | (1.91) | (0.73) | (0.59) |
| Mean of outcome (%) | 77.3 | 18.4 | 1.9 | 41.2 | 7.6 | 5.2 |
| Observations | 474,271 | 475,389 | 344,150 | 475,389 | 475,389 | 475,389 |
| Cells | 3,986 | 3,979 | 3,271 | 3,979 | 3,979 | 3,979 |
| Countries | 32 | 32 | 27 | 32 | 32 | 32 |
| *Panel C: Women* | | | | | | |
| Suitable × lagged price | −1.43 | 1.40 | 0.53 | −4.85** | 3.84*** | −1.06* |
| | (1.70) | (0.88) | (0.41) | (1.94) | (1.35) | (0.61) |
| Mean of outcome (%) | 54.7 | 7.0 | 0.5 | 27.0 | 16.5 | 3.8 |
| Observations | 1,217,450 | 1,182,257 | 772,198 | 1,182,257 | 1,182,257 | 1,182,257 |
| Cells | 4,119 | 4,119 | 3,033 | 4,119 | 4,119 | 4,119 |
| Countries | 33 | 33 | 27 | 33 | 33 | 33 |

*Notes*: Each entry reports equation (1) for an indicator equal to 100 if the respondent's own occupation is in the sector and 0 otherwise, so that coefficients are in percentage points, in all countries with surveys up to 2018. The coefficient is the change in a fully gold-suitable cell, relative to a cell with no suitable land, as the lagged gold price index (2013 equal to one) rises by one; the index rose by 0.87 between 2000–05 and the peak, so 0.87 times a coefficient is the effect of the boom (divide by 100 for the units of Girard et al. (2025)). Currently working (column 1) equals 100 if the respondent reports working at the time of the interview (DHS variables `v714` for women and `mv714` for men) and 0 if not; a respondent who worked in the past 12 months but not at the interview counts as 0. Panel A stacks the men's and women's interviews and controls for sex. Extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Cells and countries count those with respondents in each regression. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

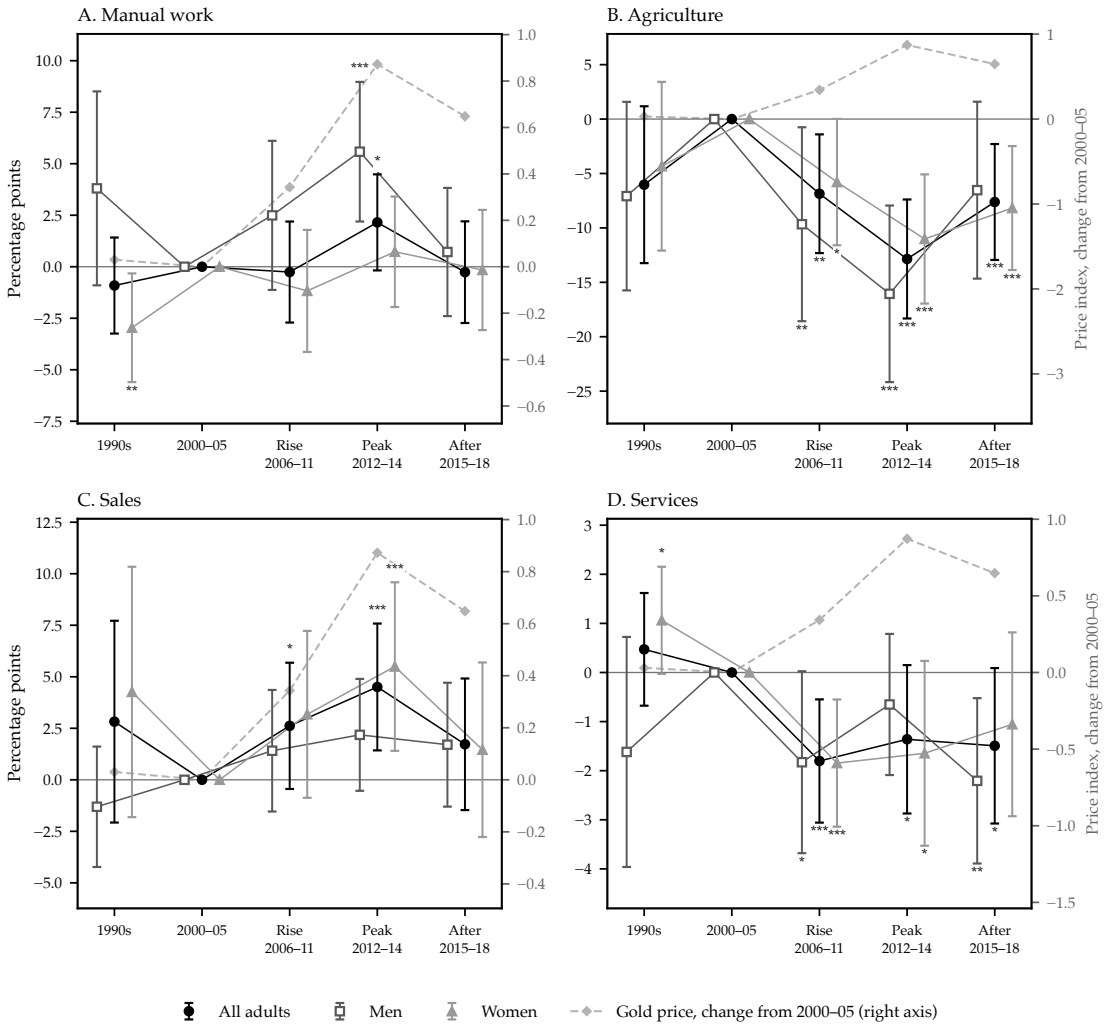



Figure 3—Work over the gold price cycle

*Notes*: Coefficients of equation (2) for manual work (Panel A), agriculture (Panel B), sales (Panel C) and services (Panel D), for all adults, men and women, each respondent's own occupation as an indicator equal to 100 or 0, so that coefficients are in percentage points, in the eight countries surveyed both at the peak and after it, surveys to 2018, relative to 2000–05; Appendix Table B1 reports the estimates. The dashed line (right axis) is the mean lagged gold price index in each window minus its mean in 2000–05, the trough of the price (2013 equal to one): 0.03 in the 1990s, 0.34 during the rise, 0.87 at the peak and 0.65 after it; each panel scales the right axis so that its zero lies on the zero line of the estimates. Section 5.2 reports the second rise. Controls and effects as in Appendix Table B1: temperature, the crop-price index, age and age squared (and sex for all adults), with cell and country-by-year effects. Data from the Demographic and Health Surveys (ICF, 2026). Bars are 95 percent confidence intervals from standard errors clustered by cell. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

level before the boom and farming has recovered more than half of its fall. Men return mostly to farming for part of the year. In the surveys that ask whether a man works all year or seasonally, his part-year farming rises by 8.6 points after the peak, while all-year farming changes by 1.2, consistent with men who combine farming with seasonal work off the farm.[4]

[4]Shares of all men with an occupation, in the 26 surveys of the eight countries that ask most working men whether they work all year, seasonally or occasionally; part of the year is seasonal or occasional. At

Only part of the manual response is recorded as mining. In the seven countries whose occupation codes name mining or quarrying, men's manual work rises by 6.9 points at the peak and falls by 5.3 after it, while extraction rises by 1.2 points on a mean of 1.3 (Appendix Table B1, column 3, and Appendix Table B8). Extraction nearly doubles, yet it is about a fifth of the manual response; the rest appears under other manual labels, consistent with diggers whom the surveys record as manual workers (Section 2).

Women follow a different path. They leave farming at the peak, by 11.0 points on a mean of 25, but the jobs they take are in sales (Panel C), up 5.5 points at the peak and down 4.0 after it, and their manual work does not change. Services fall during the boom, by 1.4 points at the peak among all adults on a mean of 4.2, imprecisely (Panel D). The share of women in any work falls as well. It is no different in the 1990s, falls by 7.5 points during the rise and by 8.1 points at the peak, 14 percent of its mean of 57, and recovers only 2.1 of those points after the peak, an estimate whose 95 percent interval reaches 5.9. I return to women's work in the second rise in Section 5.2.[5]

The 1990s coefficients are mostly imprecise. Farming in gold-suitable cells was 4 to 7 points lower than in 2000–05, and women's manual work 3 points lower. In both halves of the decade the lagged price index was within 0.04 of its mean in 2000–05, so I also split the 1990s into the surveys of 1990–95 and those of 1996–99.[6] None of the 12 estimates for all adults' farming and manual work, men's farming and manual work and women's sales and current work is statistically significant at the 5 percent level. In neither half of the decade was farming more than 3 points higher than in 2000–05, so gold-suitable places were not already leaving farming before the price rose. Section 6 pools the 1990s with 2000–05 as the reference. The responses to the fall compare the peak with the years after it, so they do not depend on the reference period.

The censuses do not show the move out of farming. In Online Appendix OA4, I estimate the census version of equation (1), equation (5), on the IPUMS-International censuses of 13 countries. Men's employment in mining rises with the price, by 0.8 points per unit of the price index on a mean of 1.8, but men's share in agriculture does not fall: it rises by 1.3 points, imprecisely, on a mean of 42, and that of all workers by 2.8, where the surveys show a fall. The two sources measure different things. No country has a census both at the peak and after it, so the censuses cannot follow the cycle, and their units are districts or regions, several times larger than a cell, in which gold-suitable land is a smaller share of where people live. I therefore use the censuses

the peak, all-year farming falls by 10.9 points and part-year farming by 6.4, both imprecisely, and part-year manual work rises by 2.9 points and falls by 2.8 after the peak.

[5] Byambasuren and Nguyen (2026) find that women's employment in mining and in retail rises with the gold price. They measure employment by industry, for ever-married women and the husbands they report, while I measure each respondent's own occupation and current work; both patterns can hold if the women who keep working move toward mining and trade while others stop working.

[6] The early surveys are Nigeria 1990, Senegal 1992 and 1993 and Mali 1995; the late ones are Benin and Mali 1996, Senegal 1997 and Guinea 1999. Men's farming is 2.9 points higher in 1990–95 and 8.7 points lower in 1996–99 than in 2000–05, and their manual work is 5.6 and 3.5 points higher.

as a check on the mining response, which is imprecise with country clusters, and not as a test of the farming result.

### 5.2 Work in the Second Rise

From 2019, the price rose again, and the surveys of 2019–25 interviewed men at a higher price than the peak surveys did: averaged over the interview dates of the men's surveys, the lagged index was 1.31, against 1.13 at the peak and 0.88 in 2015–18. Ten countries were surveyed both in 2015–18 and in 2019–25: Angola, Cameroon, Ethiopia, Malawi, Mali, Nigeria, Rwanda, Senegal, Tanzania and Zambia. For these, I estimate equation (2) on all their surveys, with age controls only, because the temperature and crop-price controls end in 2018, and I report the change between the two windows in a fully gold-suitable cell (Appendix Table B2).

Men did not leave farming again. Their farming rose by 2.0 points between the two windows, and its 95 percent interval rules out a fall of more than 2.5 points; at the first peak, in the eight countries, it had fallen by 16.1. Farming also became more seasonal: in the surveys that ask, men's part-year farming rose by 6.1 points and their all-year farming fell by 3.3. Their manual work changed by 0.1 points, with an interval from $-3.2$ to 3.5, against a rise of 5.6 at the first peak. The estimates hold without the interviews of 2020 and 2021, when COVID-19 disrupted fieldwork, and in cells surveyed in both windows, and the estimate for manual work also holds without any single country and among men aged 15 to 24 at interview, most of them children at the peak.[7] Women's sales did not rise again: they changed by $-0.4$ points and women's current work by $-1.2$, and among all adults manual work and sales stay flat.

The cycle and the second rise are measured in different countries, eight for the first and ten for the second, so the contrast could reflect which countries were surveyed. Five countries enter both groups: Mali, Nigeria, Rwanda, Senegal and Zambia. In these five alone, men's farming falls by 19.5 points at the peak and recovers 9.2 after it, men's manual work gives up 5.8 points after the peak, and women's sales give up 4.0 (Appendix Table B8, Panel D). In the second rise, men's farming in these five countries changes by 3.6 points, men's manual work by $-1.0$ and women's sales by 0.3.

The flat second rise is not confined to insecure places. In Appendix Table B7, Panel B, I split it by whether the cell had any armed conflict event in 2019–24 (Sundberg and Melander, 2013), and men's manual work is flat in both groups. The only sign of a response is in the few cells with a mine on the satellite map of Tang and Werner (2023), which hold 6 percent of respondents: there, men's manual work rises by 9.1 points and their farming falls by 12.1, both imprecisely. Ghana's suspension of small-scale mining

[7] Without 2020 and 2021, men's farming changes by 1.8 points, their manual work by 0.6 and women's sales by $-0.4$; in cells with respondents in both windows, by 3.0, 0.1 and $-0.7$. For manual work, the estimates are $-0.9$ among men aged 15 to 24, and between $-0.8$ and 1.1 when each of the ten countries is dropped in turn; the upper end of the 95 percent interval is at most 4.6 points in every case.

in 2017 (Hilson, 2017) plays no part, because Ghana is in neither the eight countries nor the ten. Other changes after 2019 could have muted the response: machines that raise output without raising employment, enforcement against illegal mining, the cost of mercury and fuel, and food prices in 2021 and 2022. The surveys measure none of them. Online Appendix OA2 compares the second rise with the change that the first boom's response implies, and splits it by earlier mining and by age; none of these comparisons settles why the response was flat.

### 5.3 Forest Clearing Does Not Slow

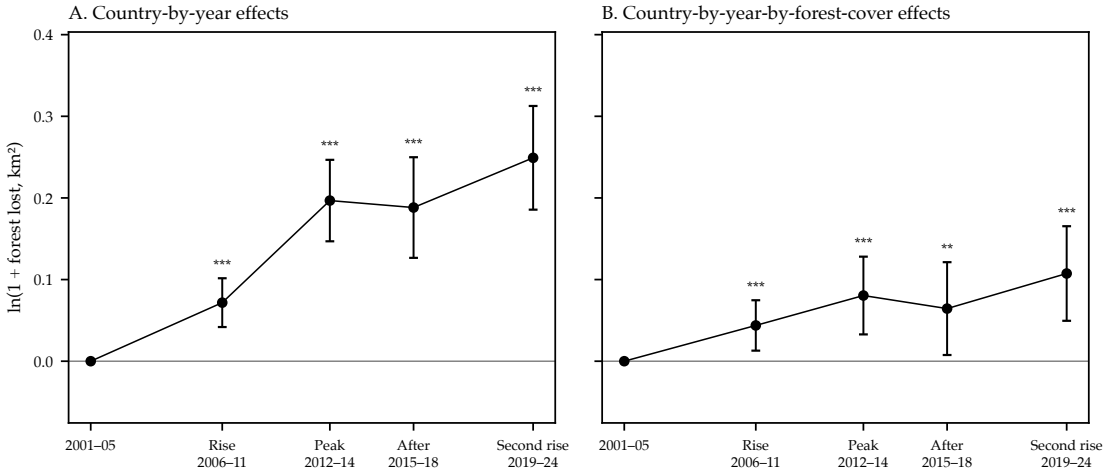



Figure 4—Forest loss over the gold price cycle

*Notes*: Coefficients of equation (2) for the logarithm of one plus the area of 2000 forest lost in the year ($km^2$), in the 4,626 African cells with at least 1 $km^2$ of forest in 2000, 2001 to 2024, relative to 2001–05, the reference period of equation (2) from the first year of the forest data. Panel A uses cell and country-by-year effects; Panel B replaces the country-by-year effects with country-by-year effects for each tercile of 2000 forest cover. Forest loss from Hansen et al. (2013), version 1.12. Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The land follows a different path from the jobs. In Figure 4, Panel A, I plot equation (2) for the area of forest lost each year in all 4,626 forested cells of the continent. During the rise, forest loss in a fully gold-suitable cell is 0.07 log points higher than in 2001–05, and at the peak 0.20 log points higher.[8] After the price falls, the excess barely changes. Clearing continues at its peak rate in the years when men return to farming, so the land cleared during the boom keeps growing. Panel B allows the improvement in satellite detection after 2012 to differ with each cell's forest cover. The coefficients are smaller, but the shape is the same: loss rises with the boom and does not fall after the peak.

In the second rise, clearing speeds up again. Forest loss in a fully gold-suitable cell

[8]The outcome is the logarithm of one plus the area lost, so 0.20 log points is a rise of 22 percent in one plus the area; at the mean of the outcome, where one plus the area is about 3 $km^2$, the area lost itself rises by about 30 percent.

grows by a further 0.06 log points between 2015–18 and 2019–24 (Panel A), and by 0.04 when detection is allowed to change with forest cover (Panel B); both differences are statistically significant at the 1 percent level. The change from the peak to 2015–18 is −0.01 log points, and its 95 percent interval excludes a fall larger than 0.044. Clearing of gold-suitable land sped up in the years when the higher price did not draw men out of farming again (Section 5.2).

A steady rise in clearing would look like this if gold-suitable land were simply being cleared faster over time. The yearly price separates the two, because it fell from 2013 to 2016 and rose again from 2019. Section 6 applies the same test to work; for forest, I estimate equation (1) with the suitable share times a linear trend added. A unit rise in the lagged price index raises forest loss in a fully gold-suitable cell by 0.17 log points with the trend included, against 0.26 without it, and by 0.12 when detection is allowed to differ with forest cover, where the trend is zero (Appendix Table B8, Panel G). Forest loss follows the price beyond a linear trend in gold-suitable land.

Table 3—Forest loss by driver and by place

| | All forested cells | | | Recorded gold or mining | | Clearing in 2001–05 | |
|---|---|---|---|---|---|---|---|
| | All loss | Mining pixels | Other pixels | Yes | No | High | Low |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Rise, 2006–11 | 0.072*** | 0.005* | 0.071*** | 0.078*** | 0.060*** | 0.048* | 0.085*** |
| | (0.015) | (0.002) | (0.015) | (0.024) | (0.021) | (0.025) | (0.019) |
| Peak, 2012–14 | 0.197*** | 0.024*** | 0.194*** | 0.237*** | 0.162*** | 0.173*** | 0.178*** |
| | (0.025) | (0.006) | (0.025) | (0.041) | (0.035) | (0.040) | (0.033) |
| After, 2015–18 | 0.188*** | 0.023*** | 0.188*** | 0.252*** | 0.144*** | 0.159*** | 0.173*** |
| | (0.031) | (0.007) | (0.031) | (0.051) | (0.043) | (0.047) | (0.041) |
| Second rise, 2019–24 | 0.249*** | 0.036*** | 0.247*** | 0.240*** | 0.232*** | 0.188*** | 0.255*** |
| | (0.032) | (0.008) | (0.032) | (0.051) | (0.045) | (0.046) | (0.044) |
| After minus peak | −0.009 | −0.001 | −0.007 | 0.016 | −0.018 | −0.014 | −0.005 |
| | (0.018) | (0.003) | (0.018) | (0.028) | (0.025) | (0.026) | (0.025) |
| Second rise minus after | 0.061*** | 0.013*** | 0.059*** | −0.012 | 0.088*** | 0.029 | 0.083*** |
| | (0.014) | (0.004) | (0.014) | (0.023) | (0.019) | (0.021) | (0.019) |
| Mean of outcome | 1.142 | 0.021 | 1.138 | 1.298 | 1.059 | 1.389 | 0.900 |
| Cell-years | 111,024 | 111,024 | 111,024 | 38,904 | 72,000 | 55,128 | 55,848 |
| Cells | 4,626 | 4,626 | 4,626 | 1,621 | 3,000 | 2,297 | 2,327 |

*Notes*: Each column reports equation (2), suitable × window, for the logarithm of one plus the area of 2000 forest lost in the year ($km^2$), in cells with at least 1 $km^2$ of forest in 2000, 2001 to 2024, relative to 2001 to 2005, with cell and country-by-year effects; coefficients and means are in log points. Forest loss is recorded from 2001, so there is no 1990s window, and the second rise ends in 2024. The 4,626 forested cells in the estimation sample lie in 49 countries; the splits in columns 4 to 7 lose cells that lack the splitting variable. Column 1 repeats Figure 4, Panel A. Columns 2 and 3 split each year's loss by the dominant driver of tree-cover loss in 2001 to 2024 mapped by Sims et al. (2025) at 1 km: column 2 counts loss in pixels whose driver is hard commodities (mining and energy infrastructure), column 3 loss in all other pixels. Columns 4 and 5 split cells by any record of gold or artisanal gold mining in the data of Girard et al. (2025) (Figure 2, Panel C); columns 6 and 7 by whether the cell's mean share of forest lost each year in 2001 to 2005 is above its country's median. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Table 3, I ask what kind of clearing this is. Sims et al. (2025) map the dominant driver of tree-cover loss over 2001 to 2024 for each square kilometre, and I split each year's loss by the driver of the pixel it falls in. Loss in pixels attributed to mining

and energy infrastructure roughly doubles at the peak in a fully gold-suitable cell, by 0.024 log points on a mean of 0.021; it does not fall after the peak and rises by a further 0.013 in the second rise (column 2). Most of the additional clearing lies in pixels attributed to other drivers, such as permanent agriculture, shifting cultivation and logging, whose loss rises by 0.19 log points at the peak and by a further 0.06 in the second rise (column 3). Girard et al. (2025) conclude that the deforestation of the boom comes more from clearing for mining itself than from indirect channels. The driver map cannot overturn that conclusion, because it assigns one driver to each pixel for the whole period and is least accurate for the rare mining class, so this loss mixes land cleared for farms and settlements around the sites with mining that the map assigns to another driver. Mining-related clearing reaches well beyond mine sites elsewhere: in the Brazilian Amazon, deforestation linked to mining extends up to 70 kilometres beyond mining leases (Sonter et al., 2017), and in the Peruvian Amazon artisanal gold mining cleared forest at rising rates as the gold price climbed (Asner et al., 2013). The second-rise clearing lies away from the mines. It rises by 0.09 log points in cells with no recorded history of gold mining and not at all in cells with such a record, although the record is undated and the driver map assigns each pixel one driver for the whole period, and it is concentrated in cells that cleared little forest in 2001–05 (columns 4 to 7). An expanding agricultural frontier would produce the same pattern, and the crop-price control ends in 2018, before cocoa prices rose in 2023 and 2024, so I do not attribute the clearing of the second rise to mining.[9]

The household surveys record no mining equipment. In Online Appendix OA5, I relate each African country's imports of excavators, other mining machines and mercury to the gold-suitable share of its land. Excavator imports rise with the boom and stay high after it, but imports of road rollers, a construction machine, also rise after the peak, and mercury imports do not rise again in the second rise, so the trade data do not separate machines for artisanal mining from a wider construction boom.

### 5.4 Human Development over the Cycle

Men move between farming and manual work as the price moves, and women move out of farming and out of work (Section 5.1). For schooling, living standards and health, the cycle leaves a longer trace. In Figure 5, I follow living standards, household wealth, night lights and child mortality through the same windows; Appendix Table B4 reports these estimates together with school attendance and three household amenities. Living standards and wealth come from households in the eight countries surveyed on both sides of the peak, and night lights and mortality from all countries with data.

[9] On the satellite map of mine footprints of Tang and Werner (2023), the increase from 2015–18 to 2019–24 is 0.066 log points (standard error 0.015) in the 4,366 forested cells without a mapped mine and 0.004 (0.060) in the 254 cells with one; the two groups hold 4,620 of the 4,626 cells, because a cell that is the only one of its group in its country is absorbed by the country-by-year effects.

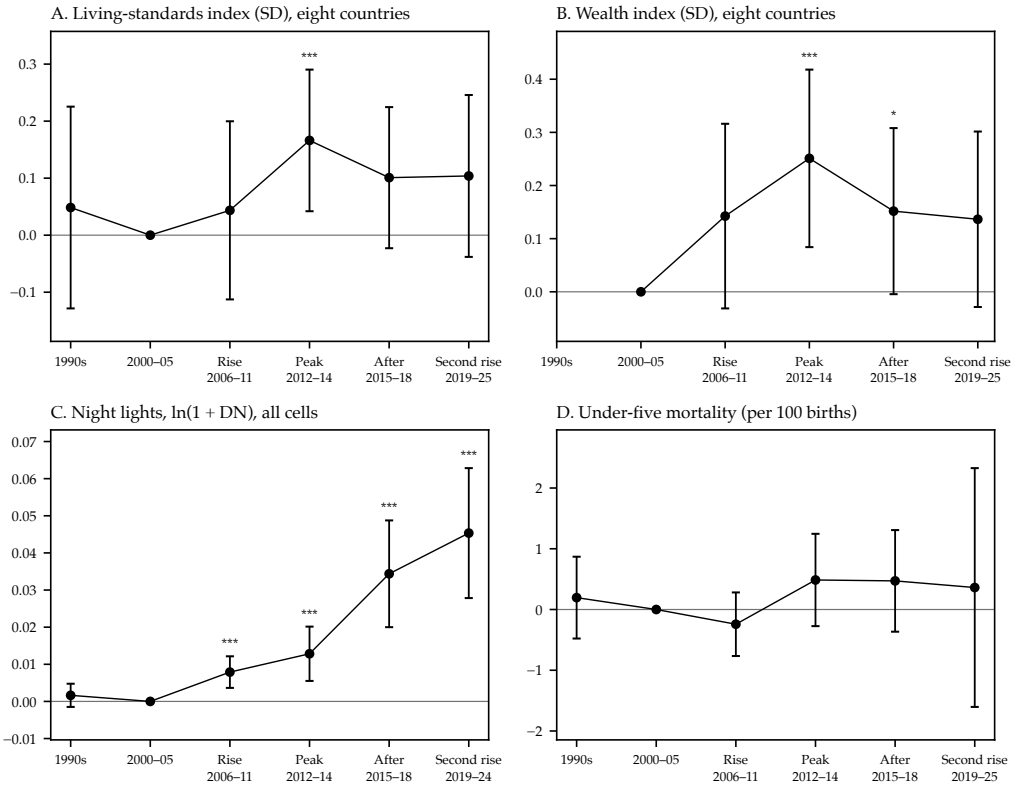



Figure 5—Living standards and health over the gold price cycle

*Notes*: Coefficients of equation (2) in a fully gold-suitable cell, relative to 2000–05. Panel A is the living-standards index, the mean of the wealth index, electricity, improved water and (negatively) no toilet facility, each standardised on the households of 2000–05; Panel B is the wealth index, standardised within each survey; both use households in the eight countries surveyed both at the peak and after it. Panel C is the logarithm of one plus mean harmonised night lights (DN, the digital number of the harmonised series) in all African cells, 1992 to 2024, so its last window ends in 2024. Panel D is under-five mortality per 100 births, by year of birth, with cell-by-survey and country-by-birth-year effects; Panel C uses country-by-year effects for each brightness group before the boom, in 1992–2005. Household and birth data from the Demographic and Health Surveys (ICF, 2026); night lights from Li et al. (2020). Appendix Table B4 reports the estimates, the specifications and Holm-adjusted $p$-values. Bars are 95 percent confidence intervals from standard errors clustered by cell. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

For teenagers, the comparison can be made within a place. Children aged 6 to 11 in the same cell and survey would also feel a school closure, a mining camp or mercury, but they are too young for the pits. I therefore estimate

$$D_{gcs} = \sum_{w \in \mathcal{W}} \beta_{wg} \left( S_c \times \mathbf{1}[t \in w] \right) + \alpha_{cg} + \lambda_{ktg} + u_{gcs}, \tag{3}$$

where $D_{gcs}$ is the attendance rate of age group $g$ (12 to 15, 16 to 20 or 21 to 24) minus that of children aged 6 to 11 in cell $c$ and survey $s$, $t$ is the year of survey $s$ and $k$ the country of cell $c$, $\alpha_{cg}$ are cell-by-age-group effects, $\lambda_{ktg}$ country-by-year-by-age-group effects and $u_{gcs}$ the error. Each observation is the mean of an age group in a cell and survey, weighted by the number of household members in the group, and the estimates start in 2000, because only one earlier survey records attendance.

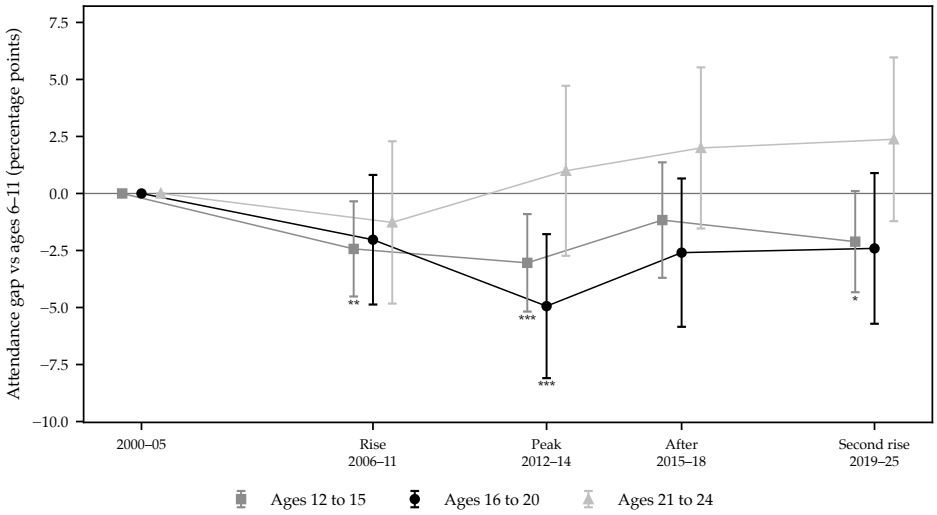



Figure 6—School attendance by age over the gold price cycle

*Notes*: Coefficients of equation (3): the attendance rate of each age group minus that of children aged 6 to 11 in the same cell and survey, in a fully gold-suitable cell, relative to 2000–05, in percentage points (attendance is an indicator equal to 100 or 0). All countries with household rosters recording attendance, 2000 to 2025; Each observation is an age group in a cell and survey, weighted by the number of household members in the group. Appendix Table B5, Panel A, reports the estimates. Data from the Demographic and Health Surveys (ICF, 2026). Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In Figure 6, I plot the result for each age group. At the peak, attendance at ages 16 to 20 in a fully gold-suitable cell is 4.9 points lower than among younger children in the same place, against an attendance rate of 47 percent at those ages, and at ages 12 to 15 it is 3.0 points lower. Young adults aged 21 to 24, who have mostly left school, show no change, so the fall is specific to school-age teenagers. Boys and girls lose attendance alike (Figure 7, Panel A). Measured against younger children in the same place, the fall appears both where the household head has no schooling and where the head has some, at 6.2 and 4.3 points. Young men aged 15 to 19 leave farming at the peak, by 18 points on a mean of 32, and return to it after the peak, as older men do (Appendix Table B8, Panel G). Attendance returns only in part, imprecisely. The gap at ages 16 to 20 narrows by about half after the peak, to 2.6 points, and is 2.4 points in the second rise (Appendix Table B4, column 1), but neither later gap is precisely estimated, so the data do not tell a full return from none. Completed schooling shows no detectable loss. Among household members aged 21 to 45, I regress years of schooling on the suitable share of the cell times the mean lagged price index in the years in which the person's birth cohort was aged 12 to 20, with industrial gold mines times the same price, so that cohorts near industrial mines do not drive the estimate, age, its square and sex, and with cell-by-survey and country-by-birth-year effects. Years of schooling do not fall with that price: the estimate is $-0.14$ years on a mean of 5.9, with a 95 percent interval from $-0.36$ to 0.09 (Appendix Table B8, Panel G). Counting a year of attendance as a year of schooling, the gaps at the peak, 3.0 points at ages 12 to 15 and 4.9 at ages 16 to

20, would remove about 0.37 years of schooling from a cohort that faced them through its teenage years, or 0.43 years per unit of the index, which the interval excludes. The attendance gaps pooled over all windows imply a smaller loss, about 0.19 years per unit of the index, and the smallest loss this test detects with 80 percent power is 0.32 years, so it cannot rule out that smaller loss. Those born in 1990 to 1999, teenagers during the boom, have as much schooling relative to other cells as those born in 1985 to 1989.[10] Ghazarian and Khan (2025) find lower completed schooling among adults aged 21 to 30 who faced a higher gold price in childhood and adolescence, most of all at ages 6 to 10. Their design differs from this test in three ways: exposure is the distance to gold-suitable land, places are compared within regions, where this test compares them within cells, and the price at ages 6 to 10, which carries most of their effect, is not part of the teenage price used here. Because schooling in gold-suitable cells has fallen behind that of other cells since the cohorts of the 1960s, a comparison of cohorts without cell effects can also attribute part of that trend to the price. The persistence I find is therefore a feature of places: teenagers in gold-suitable cells attend school less after the peak than before the boom.

Households in gold-suitable places rise in their country's wealth ranking and stay there. The wealth index, built from assets and housing and standardised within each survey, rises by 0.25 standard deviations at the peak in a fully gold-suitable cell (Figure 5, Panel B); among rural households alone it rises by 0.17, so part of the gain may come with a shift toward towns.[11] When the price falls, it gives up 0.10 standard deviations, two fifths of its gain, and it holds the rest through 2019–25; the second rise leaves it at that level. With the yearly price and a linear trend, wealth rises by 0.32 standard deviations per unit of the index and the trend is negative, and the attendance gap at ages 16 to 20 widens by 6.3 points per unit, against 2.5 without the trend, with a trend of the opposite sign. The windows themselves cannot be estimated precisely together with the trend, so the data do not separate persistence from a trend that began with the boom (Appendix Table B3).

Night lights tell the same story at the scale of the cell (Panel C); the peak estimate does not depend on the change of satellite sensor in 2014, since the years before the change alone give 0.012 log points against 0.013 in the full series (Appendix Table B8, Panel G), although the later rise rests on the newer sensor. They brighten during the boom, and instead of dimming when the price falls they keep rising, so that gold-suitable cells are brighter after the peak and in the second rise than at the peak itself. Relative to other cells, gold-suitable cells were no brighter in the 1990s than in 2000–05,

[10] The cell is the place of residence at the interview, not in the teenage years. Schooling in gold-suitable cells has fallen behind that of other cells since the cohorts born in the 1960s, by 0.08 years per decade of birth, and no cohort of the boom departs from that trend; with the trend included, the estimate for the teenage price is 0.07 years.

[11] With country clusters, the wild cluster bootstrap $p$-value is 0.20 for the wealth index at the peak and 0.08 for the living-standards index; dropping each country in turn leaves the wealth estimate between 0.18 and 0.28.

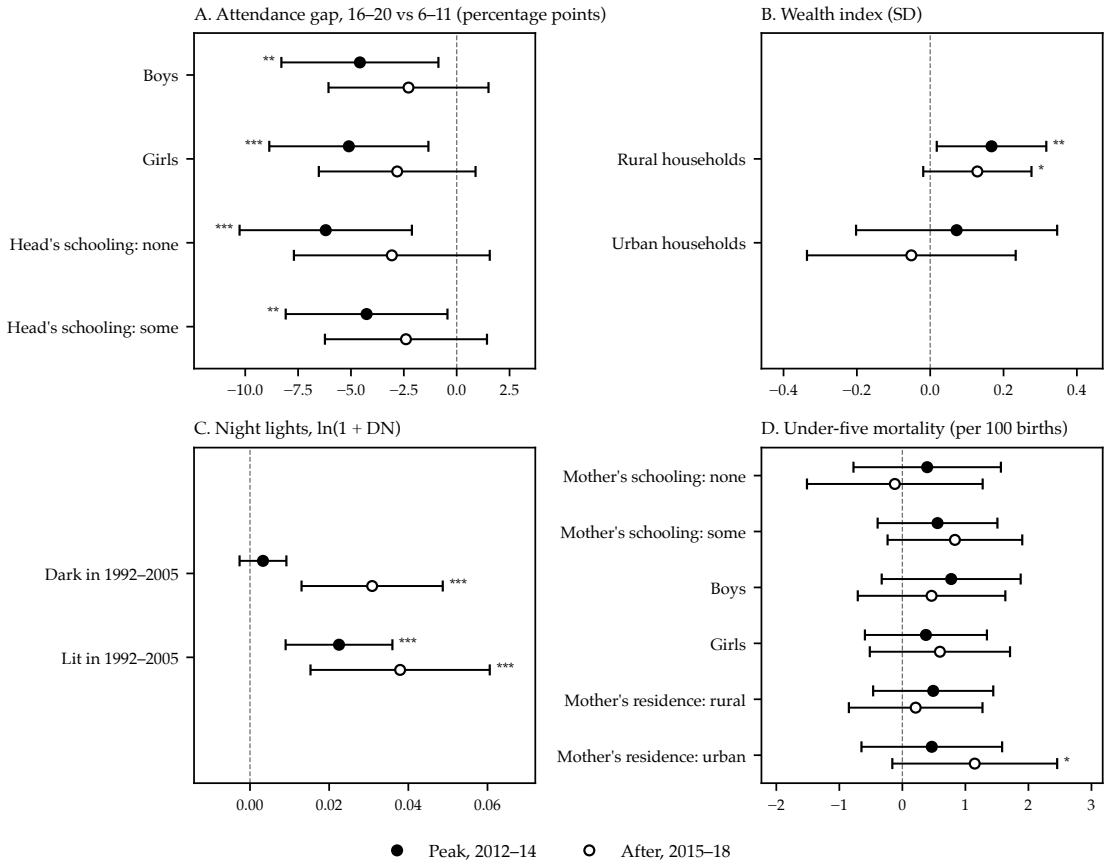



Figure 7—Human development over the gold price cycle, by group

*Notes*: Coefficients of equation (2) at the peak (2012–14, filled) and after it (2015–18, hollow), relative to 2000–05, estimated separately for each group with the specifications of Appendix Table B4: the attendance gap of equation (3) by sex and by the schooling of the household head (Panel A), the wealth index for rural and urban households in the eight countries surveyed both at the peak and after it (Panel B), night lights for cells dark and lit in 1992–2005 (Panel C), and under-five mortality by the mother's schooling, the child's sex and the mother's urban or rural residence at interview (Panel D); Appendix Table B5, Panels B to E, reports the estimates. Bars are 95 percent confidence intervals from standard errors clustered by cell. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

so the rise does not continue an earlier trend, but from 2006 on a linear trend for gold-suitable cells fits it about as well as the windows do: with the trend, the peak coefficient is zero, and lights rise with the yearly price by 0.017 log points per unit of the index, against 0.040 without it (Appendix Table B3).

The household amenities move less cleanly. Electricity and improved drinking water are both higher at the peak than in 2000–05 (Appendix Table B4, columns 6 and 7), but each already differed in the 1990s, electricity in the same direction and water in the opposite one, so I cannot attribute either to the boom. The share of households without a toilet facility does not change at the peak and falls by 3.9 points from 2015–18 to the second rise (column 8). Taken together, the index of the four rises by 0.17 standard deviations at the peak and stays 0.10 above its level of 2000–05 after it (Figure 5, Panel A); its fall after the peak, 0.07, is imprecise. Under-five mortality does not respond

in any window (Panel D): the estimates are small relative to a mean of 11 deaths per 100 births. After a Holm adjustment for the four main outcomes (attendance, night lights, mortality and the living-standards index; Holm, 1979), the peak estimates for attendance, night lights and the living-standards index remain statistically significant at the 5 percent level and that for mortality does not;[12] the changes after the peak in attendance, mortality and the living-standards index do not survive the adjustment, while the continued rise in night lights does. Child deaths rise only in towns (Figure 7).

In Figure 7, I split these outcomes by group, without adjusting for the number of splits. Wealth rises by 0.17 standard deviations at the peak among rural households and by 0.07, imprecisely, among urban ones (Panel B). Both are smaller than the 0.25 for all households. The share of surveyed households in clusters classified as urban rises by 9 points at the peak, imprecisely, on a mean of 36 percent, consistent with the small towns that artisanal gold created (Girard and Pignède, 2026). Light comes first to places that were already lit (Panel C). Cells lit before the boom, in 1992–2005, brighten at the peak and again after it, while cells that were dark then light up only after the peak. At the peak, under-five mortality does not respond for any group, whether split by the mother's schooling, the child's sex or residence (Panel D). After the peak, it is 1.1 points higher among children of urban mothers, on a mean of 8.4 deaths per 100 births, and 0.2 points higher among children of rural mothers. Residence at interview moves with the boom, so in Appendix Table B10, I instead split cells by whether they held an urban cluster in a survey fielded before 2006, close to the towns of 1975 that Girard and Pignède (2026) follow. In these towns, under-five mortality is 1.0 points higher at the peak and 1.2 points higher after it, on a mean of 10.5, and in the other cells it does not change. As in Girard and Pignède (2026), child mortality rises with artisanal gold in these towns and not in other cells, although this is one of several splits by group and holds less clearly with towns defined by night lights.[13]

## 6 Robustness

In Figure 8, I re-estimate the cycle for the four outcomes that carry Section 5.1 under 16 changes to the specification, and Appendix Table B6 reports the estimates. The first row repeats Appendix Table B1. Measuring exposure as any gold-suitable land in the cell, as a suitable share of at least one half, or as the share of the cell within 10 kilometres of suitable land rescales the coefficients but leaves the cycle in place. Controlling for industrial gold mines times the price, for industrial mines of other minerals valued at their own world prices, or for armed conflict in the cell, which mineral prices and resource

[12] The Holm-adjusted $p$-values at the peak are 0.01 for attendance, below 0.01 for night lights, 0.02 for the living-standards index and 0.21 for mortality.

[13] The rise is smaller and imprecise for deaths before age one: 0.6 points at the peak and 0.8 after it in the towns, on a mean of 6.5. Defining towns as cells lit at night in 1992 to 1999 gives 0.7 points at the peak, imprecisely, and 1.0 after it, but that definition counts 83 percent of births as urban.

rents can fuel (Bazzi and Blattman, 2014; Berman and Couttenier, 2015; Berman et al., 2017; Dube and Vargas, 2013; Sánchez de la Sierra, 2020; Couttenier et al., 2017), moves no estimate of the farming cycle by more than 1.0 point, and neither does dropping the temperature and crop controls; conflict events fall in gold-suitable cells of the eight countries at and after the peak (Appendix Table B8, Panel G). Rainfall, which follows geography and could move farming in the same places, does not explain the cycle. With CHIRPS rainfall in the cell in the interview year and the year before (Funk et al., 2015), its interaction with the suitable share, or indicators for drought years, men's farming falls by 15.0 to 15.5 points at the peak and recovers 9.2 to 11.2.[14] Shocks common to farming zones do not explain it. With country-by-zone-by-year effects, where a zone combines the length of the growing season with the cell's main crop, men's farming falls by 20.7 points at the peak and recovers 8.3; with secondary diamonds and gem deposits, mined artisanally as well, interacted with each window, it falls by 15.4 and recovers 8.7.[15] Pooling 1990–2005 as the reference period moves the peak coefficients toward zero, to 4.5 points for men's manual work, −13.9 for their farming and 4.0 for women's sales, because the 1990s already differed, but it leaves every change after the peak within 0.1 points.

Changes in who was surveyed do not explain the cycle. The urban share of surveyed households in gold-suitable cells rises by 9 points at the peak, on a mean of 36 percent (Appendix Table B8, Panel G), and farming is less common in towns. In rural clusters alone, men's farming falls by 14.0 points at the peak and recovers 12.0. The cycle keeps its shape, if smaller, with urban residence interacted with each window, in cells with respondents in every window, and without the clusters whose displacement could have moved them across a cell edge, where men's farming recovers 6.2 to 7.9 points. With other mines, conflict and urban residence controlled together, men's farming falls by 12.8 points and recovers 7.8, and their manual work rises by 4.9 and gives up 4.3. Nor does the coding of occupations move with the cycle.[16]

Arrivals at the peak do not produce the cycle in work, although this check cannot speak to people who left. Among respondents who have always lived in the place or have lived there for at least five years, men's manual work rises by 8.2 points at the

[14] Rainfall is the annual total in the cell standardised by its 1981–2010 mean and standard deviation, and drought is rainfall more than one standard deviation below the mean. Across the three specifications, men's manual work rises by 5.5 to 5.8 points at the peak and gives up 4.6 to 4.9, women's sales rise by 4.5 to 4.9 and give up 3.7 to 4.7, and men's farming changes by 1.0 to 2.2 points in the second rise, against 2.3 without rainfall in the same sample.

[15] Growing season, main crop and mineral deposits from the PRIO-GRID static variables (Tollefsen et al., 2012). With the zone effects, men's manual work rises by 5.8 points at the peak and gives up 4.0, and women's sales rise by 4.4 and give up 2.5, imprecisely, as with region-by-year effects. In the second rise, men's farming changes by 2.6 and 2.3 points in the two specifications and men's manual work by −1.6 and 0.6.

[16] In the eight countries, with age controls and cell and country-by-year effects, the share of men whose occupation is missing changes by −0.8 points at the peak and by 1.1 after it, on a mean of 4.9, and the share coded as another occupation by 0.4 and 1.1, on a mean of 1.2, against −15.8 and 9.3 for farming in the same sample. The share of men who report no work rises by 6.3 points at the peak and falls by 3.4 after it, on a mean of 16, both imprecisely.

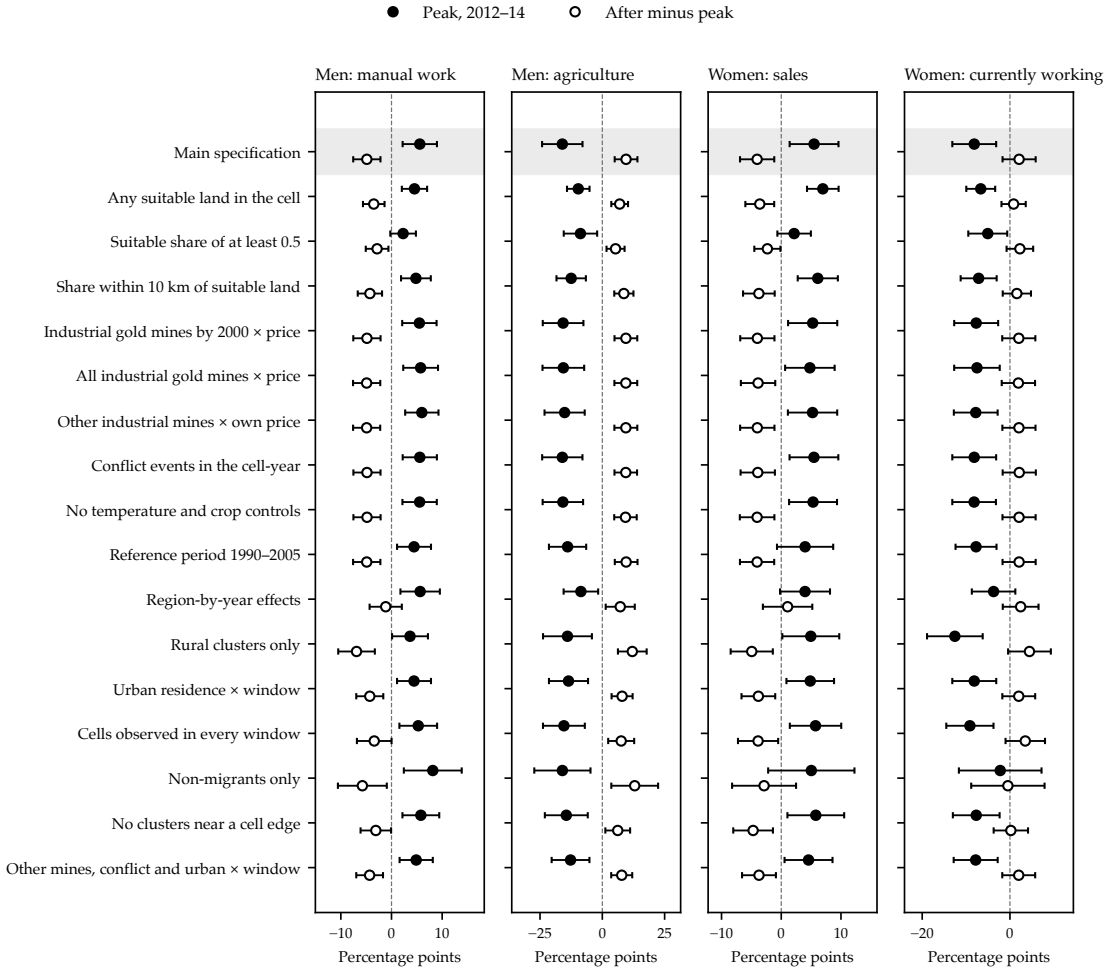



Figure 8—Robustness of the cycle

*Notes*: Each row reports equation (2) for the outcome in the panel title, in percentage points, in the eight countries surveyed both at the peak and after it, surveys to 2018, with the change named in the row: the coefficient at the peak, 2012–14, relative to 2000–05 (filled), and the change from the peak to 2015–18 (hollow). The first row is the specification of Appendix Table B1. The shaded row is the main specification. Appendix Table B6 defines each row and reports the estimates, the range when one country is dropped, and wild cluster bootstrap and placebo $p$-values. Bars are 95 percent confidence intervals from standard errors clustered by cell.

peak and falls by 5.8 after it, and men's farming falls by 16.0 points and recovers 13.0. In Appendix Table B9, I estimate equation (2) for who lives in the surveyed households. At the peak, households in gold-suitable cells are larger by 0.4 people on a mean of 5.9, the share of young men is 1.0 point higher and the share of women 0.8 points lower, and none of these changes after the peak. Households in gold-suitable cells were also 1.4 people larger in the 1990s than in 2000–05, so household size moved before the boom as well. Women who arrived in the previous five years are no more common at the peak; they are 6.5 points more common after it, when manual work had already fallen. The one outcome that differs among stayers is women's current work, whose fall at the peak is 2.2 points against 8.1 in the full sample, so part of women's withdrawal from work may come from women who moved.

Fieldwork timing does not explain the farming cycle. Assigning each survey to the window of its first interview year, which places the 2014–15 surveys of Chad and Rwanda wholly in the peak, leaves the cycle as large: men's farming falls by 16.5 points and recovers 11.0. With country-by-year-by-interview-month effects and the suitable share times calendar-month indicators, men's farming still falls by 12.5 points and recovers 8.1, while the fall in manual work after the peak shrinks to 2.5 points and that in women's sales to 2.7, both imprecisely (Appendix Table B8, Panel E).

Two checks weaken the reversal in manual work and sales further. With region-by-year effects, which compare each cell only with others in the same survey region, men's manual work still rises by 5.7 points at the peak and men's farming still falls and recovers, by 8.6 and 7.2 points, but manual work falls by only 1.1 points after the peak and women's sales no longer fall. A placebo that shifts the whole suitability map across the continent tells the same story. Shifted maps keep the regional structure of geology. Across about 320 shifts of at least two degrees, a cycle in men's farming as large as the actual one appears in 5 percent of the shifts at the peak and 7 percent after it, and cycles in manual work and sales in 13 to 32 percent. Part of the reversal in manual work and sales is therefore common to whole regions, while the reversal in farming holds within them.

A differential trend does not explain the cycle. Adding the suitable share times a linear trend to the windows is not informative, because in eight countries the windows and the trend are almost collinear: the trend has an $R^2$ of 0.99 on the window regressors and the suitable share. The yearly price separates them, because it fell from 2013 to 2016. In Appendix Table B6, Panel C, the suitable share times the lagged price index lowers men's farming by 11.8 points per unit of the index, and by 21.9 with a linear trend for gold-suitable cells, the suitable share times decades since 2000; men's manual work and women's sales rise with the price in both specifications.

Dropping any one of the eight countries leaves the rise of men's manual work at the peak between 4.8 and 7.3 points and its fall after the peak between 3.2 and 5.7 points; farming recovers by 7.9 to 10.6 points in every sample. Standard errors that allow correlation across cells up to 100 or 500 kilometres apart (Conley, 1999) leave each of the eight cycle statistics of Appendix Table B6 that is significant with cell-clustered errors significant at the 5 percent level, except the fall of women's sales after the peak at 500 kilometres, and across the 24 cycle statistics for all adults, men and women in Appendix Table B1, a Holm adjustment leaves men's farming and manual work at the peak and after it below 0.05. With eight countries, inference across countries is weak, because survey rounds within a country differ by more than sampling error: between consecutive surveys of the same country fielded at nearly the same price, the estimate for men's farming varies by about 15 points more than sampling error alone would produce.[17] The placebo does not depend on the number of countries. In 1,000

[17] The 29 pairs of consecutive surveys whose mean lagged price differs by less than 0.15 give estimates of

reassignments of the suitable share across cells within each country, estimated on cell-by-year means,, no more than 2 percent of draws give a rise at the peak or a reversal after it as large as the actual estimates, for men's manual work and farming and for women's sales, and no more than 0.1 percent give a fall in women's current work as large as the actual one (Appendix Table B6).

The cycle does not depend on restricting the sample to the eight countries surveyed on both sides of the peak. Estimated on every country, with surveys to 2018, men's farming falls by 6.4 points at the peak and recovers 6.1, their manual work rises by 2.6 and gives up 3.1, and women's sales rise by 4.1 and give up 3.1 (Appendix Table B7, Panel A). The estimates are 40 to 80 percent as large, because different countries identify each window, but every one has the same sign and is statistically significant at the 5 percent level.

Online Appendix OA3 re-estimates Table 2 and the cycle with exposure measured as the share of the cell within 10 kilometres of gold-suitable land (see also Ghazarian and Khan, 2025), and the estimates change little: the cycle in men's manual work and women's sales is within a point of the main estimates, and the fall in men's farming at the peak is smaller, 12.5 points against 16.1.

## 7 Conclusion

Between 2000 and 2012, the gold price rose sixfold; by 2015 it had fallen by 30 percent, and from 2019 it rose above its earlier peak. I follow work, land and human development in Africa's gold-suitable places through this cycle, using 132 household surveys linked to the geology that artisanal mining needs. Men's work followed the price. In the eight countries surveyed on both sides of the peak, men in a fully gold-suitable cell took up manual work and left farming as the price rose, and farming recovered when the price fell; women moved into sales and back out of them, and the share of women in any work fell, by less among long-time residents. When the price rose again after 2019, men did not leave farming again, while forest loss grew further.

Other changes outlast the price. Forest clearing in gold-suitable cells did not slow when the price fell, so the land taken by the boom kept growing. Households kept about three-fifths of their rise in the wealth ranking, part of it perhaps from a shift toward towns, and the night lights of gold-suitable cells kept rising after the peak, along a path that a trend for these cells fits as well. Teenagers aged 16 to 20 lost 4.9

the suitable share times the later survey with a standard deviation of 17.7 points, against a mean standard error of 9.2. Wild cluster restricted bootstrap $p$-values with country clusters, over all 256 Rademacher draws, are 0.02 for the fall of men's farming at the peak and 0.08 for the rise of women's sales, and between 0.12 and 0.51 for the other six statistics in Appendix Table B6. With Webb six-point weights and 9,999 draws, the corresponding $p$-values are 0.02 for the fall of men's farming at the peak and between 0.06 and 0.54 for the other seven. A jackknife (CV3) over the eight leave-one-country-out estimates (MacKinnon et al., 2023), with $t$ critical values on seven degrees of freedom, gives $p$-values of 0.02 at the peak and 0.01 after it for men's farming, 0.04 and 0.05 for their manual work, and 0.07 and 0.17 for women's sales.

points of school attendance relative to younger children in the same place, and the gap narrowed only in part, and imprecisely, after the jobs had gone, while the completed schooling of the boom cohorts shows no detectable loss. In cells that held towns before 2006, child mortality was about 1 point per 100 births higher at and after the peak, in one of several splits by group.

The contrast between work and land matters for how artisanal mining is judged. Employment at the peak is a poor measure of what a gold rush leaves behind: the move out of farming partly reversed within a few years, and the cleared forest, the lower school attendance of gold-suitable places and the assets households bought remain. The second rise cleared more forest, much of it on land with no earlier record of mining, and drew no workers out of farming. These estimates describe gold-suitable places, not the people who mined in them, and each comparison across the cycle rests on fewer than a dozen countries. Why the second rush drew no workers out of farming is the open question. The flat response is not confined to places with a recorded history of gold mining, and men who were children at the peak responded no more than older men. The household surveys record neither equipment nor the scale of operations, the trade data cannot place a machine at a site, and the new clearing lies away from mapped mines. Repeated satellite maps of mining pits, or licence registers, would tell whether the second rush was mined by machines, by fewer people, or elsewhere, and whether enforcement shaped it.

## References


**Allcott, Hunt and Daniel Keniston**, "Dutch Disease or Agglomeration? The Local Economic Effects of Natural Resource Booms in Modern America," *Review of Economic Studies*, 2018, *85* (2), 695–731.

**Anderson, Michael L.**, "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, 2008, *103* (484), 1481–1495.

**Aragón, Fernando M. and Juan Pablo Rud**, "Natural Resources and Local Communities: Evidence from a Peruvian Gold Mine," *American Economic Journal: Economic Policy*, 2013, *5* (2), 1–25.

**_ and _**, "Polluting Industries: Evidence from Mining in Ghana," *Economic Journal*, 2016, *126* (597), 1980–2011.

**_ , _ , and Gerhard Toews**, "Resource Shocks, Employment, and Gender: Evidence from the Collapse of the UK Coal Industry," *Labour Economics*, 2018, *52*, 54–67.

**Asner, Gregory P., William Llactayo, Raul Tupayachi, and Ernesto Ráez Luna**, "Elevated Rates of Gold Mining in the Amazon Revealed through High-Resolution Monitoring," *Proceedings of the National Academy of Sciences*, 2013, *110* (46), 18454–18459.

**Atkin, David**, "Endogenous Skill Acquisition and Export Manufacturing in Mexico," *American Economic Review*, 2016, *106* (8), 2046–2085.

**Bazillier, Rémi and Victoire Girard**, "The Gold Digger and the Machine: Evidence on the Distributive Effect of the Artisanal and Industrial Gold Rushes in Burkina Faso," *Journal of Development Economics*, 2020, *143*, 102411.

**Bazzi, Samuel and Christopher Blattman**, "Economic Shocks and Conflict: Evidence from Commodity Prices," *American Economic Journal: Macroeconomics*, 2014, *6* (4), 1–38.

**Benshaul-Tolonen, Anja**, "Local Industrial Shocks and Infant Mortality," *Economic Journal*, 2019, *129* (620), 1561–1592.

**Berman, Nicolas and Mathieu Couttenier**, "External Shocks, Internal Shots: The Geography of Civil Conflicts," *Review of Economics and Statistics*, 2015, *97* (4), 758–776.

**_ , _ , Dominic Rohner, and Mathias Thoenig**, "This Mine Is Mine! How Minerals Fuel Conflicts in Africa," *American Economic Review*, 2017, *107* (6), 1564–1610.

**Black, Dan, Terra McKinnish, and Seth Sanders**, "The Economic Impact of the Coal Boom and Bust," *Economic Journal*, 2005, *115* (503), 449–476.

**Burgert, Clara R., Josh Colston, Thea Roy, and Blake Zachary**, "Geographic Displacement Procedure and Georeferenced Data Release Policy for the Demographic and Health Surveys," DHS Spatial Analysis Reports 7, ICF International, Calverton, Maryland 2013.

**Byambasuren, Tsenguunjav and Anh Nguyen**, "Artisanal Mining and Intimate Partner Violence: Evidence from Sub-Saharan Africa," 2026. Working paper, June 2026.

**Cascio, Elizabeth U. and Ayushi Narayan**, "Who Needs a Fracking Education? The Educational Response to Low-Skill-Biased Technological Change," *ILR Review*, 2022, *75* (1), 56–89.

**Charles, Kerwin Kofi, Erik Hurst, and Matthew J. Notowidigdo**, "Housing Booms and Busts, Labor Market Opportunities, and College Attendance," *American Economic Review*, 2018, *108* (10), 2947–2994.

**Conley, Timothy G.**, "GMM Estimation with Cross Sectional Dependence," *Journal of Econometrics*, 1999, *92* (1), 1–45.

**Couttenier, Mathieu, Pauline Grosjean, and Marc Sangnier**, "The Wild West IS Wild: The Homicide Resource Curse," *Journal of the European Economic Association*, 2017, *15* (3), 558–585.

**Crawford, Gordon and Gabriel Botchwey**, "Conflict, Collusion and Corruption in Small-Scale Gold Mining: Chinese Miners and the State in Ghana," *Commonwealth & Comparative Politics*, 2017, *55* (4), 444–470.

**de la Sierra, Raúl Sánchez**, "On the Origins of the State: Stationary Bandits and Taxation in Eastern Congo," *Journal of Political Economy*, 2020, *128* (1), 32–74.

**Dube, Oeindrila and Juan F. Vargas**, "Commodity Price Shocks and Civil Conflict: Evidence from Colombia," *Review of Economic Studies*, 2013, *80* (4), 1384–1421.

**Duodu, Albert and Michael Danquah**, "Artisanal Mining and the Hollowing of the Formal Sector: Evidence from Ghana's Galamsey Zones," 2026. Under review.

**Edmonds, Eric V. and Nina Pavcnik**, "The Effect of Trade Liberalization on Child Labor," *Journal of International Economics*, 2005, *65* (2), 401–419.

**Feyrer, James, Erin T. Mansur, and Bruce Sacerdote**, "Geographic Dispersion of Economic Shocks: Evidence from the Fracking Revolution," *American Economic Review*, 2017, *107* (4), 1313–1334.

**Funk, Chris, Pete Peterson, Martin Landsfeld, Diego Pedreros, James Verdin, Shraddhanand Shukla, Gregory Husak, James Rowland, Laura Harrison, Andrew Hoell, and Joel Michaelsen**, "The Climate Hazards Infrared Precipitation with Stations—a New Environmental Record for Monitoring Extremes," *Scientific Data*, 2015, *2*, 150066.

**Gaulier, Guillaume and Soledad Zignago**, "BACI: International Trade Database at the Product-Level. The 1994–2007 Version," Working Paper 2010-23, CEPII 2010.

**Ghazarian, Avenia and Akib Khan**, "Economic Opportunities and Human Capital Investments: Evidence from Artisanal Gold Mining in Africa," 2025. Misum Working Paper 2023-14, version of August 14, 2025.

**Girard, Victoire and Édouard Pignède**, "Artisanal Mining and Urbanization in Africa," Working Paper 2601, NOVAFRICA 2026.

**_ , Teresa Molina-Millán, and Guillaume Vic**, "Artisanal Mining in Africa. Green for Gold?," *The Economic Journal*, 2025, *135* (672), 2578–2597.

**Hansen, Matthew C., Peter V. Potapov, Rebecca Moore et al.**, "High-Resolution Global Maps of 21st-Century Forest Cover Change," *Science*, 2013, *342* (6160), 850–853.

**Hilson, Gavin**, "Shootings and Burning Excavators: Some Rapid Reflections on the Government of Ghana's Handling of the Informal Galamsey Mining 'Menace'," *Resources Policy*, 2017, *54*, 109–116.

**_ , Abigail Hilson, and Eunice Adu-Darko**, "Chinese Participation in Ghana's Informal Gold Mining Economy: Drivers, Implications and Clarifications," *Journal of Rural Studies*, 2014, *34*, 292–303.

**Hinton, Jennifer J., Marcello M. Veiga, and Christian Beinhoff**, "Women and Artisanal Mining: Gender Roles and the Road Ahead," in Gavin M. Hilson, ed., *The Socio-Economic Impacts of Artisanal and Small-Scale Mining in Developing Countries*, Lisse: A. A. Balkema, 2003.

**Holm, Sture**, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, 1979, *6* (2), 65–70.

**Huang, Qing, Victoria Wenxin Xie, and Wei You**, "Do Resource Rents Drive Urbanization and Structural Transformation? A Global Analysis," *Journal of Development Economics*, 2026, *180*, 103709.

**ICF**, "Demographic and Health Surveys (various) [Datasets], 1986–2025," Funded by USAID. Rockville, Maryland: ICF [Distributor]. https://dhsprogram.com 2026.

**Jacobsen, Grant D. and Dominic P. Parker**, "The Economic Aftermath of Resource Booms: Evidence from Boomtowns in the American West," *The Economic Journal*, 2016, *126* (593), 1092–1128.

**Kling, Jeffrey R., Jeffrey B. Liebman, and Lawrence F. Katz**, "Experimental Analysis of Neighborhood Effects," *Econometrica*, 2007, *75* (1), 83–119.

**Kotsadam, Andreas and Anja Tolonen**, "African Mining, Gender, and Local Employment," *World Development*, 2016, *83*, 325–339.

**Kruger, Diana I.**, "Coffee Production Effects on Child Labor and Schooling in Rural Brazil," *Journal of Development Economics*, 2007, *82* (2), 448–463.

**Li, Xuecao, Yuyu Zhou, Min Zhao, and Xia Zhao**, "A Harmonized Global Nighttime Light Dataset 1992–2018," *Scientific Data*, 2020, *7*, 168.

**MacKinnon, James G., Morten Ørregaard Nielsen, and Matthew D. Webb**, "Cluster-Robust Inference: A Guide to Empirical Practice," *Journal of Econometrics*, 2023, *232* (2), 272–299.

**Michaels, Guy**, "The Long Term Consequences of Resource-Based Specialisation," *Economic Journal*, 2011, *121* (551), 31–57.

**Minnesota Population Center**, "Integrated Public Use Microdata Series, International," University of Minnesota, extract 5 2026.

**Saavedra, Santiago and Mauricio Romero**, "Local Incentives and National Tax Evasion: The Response of Illegal Mining to a Tax Reform in Colombia," *European Economic Review*, 2021, *138*, 103843.

**Shah, Manisha and Bryce Millett Steinberg**, "Drought of Opportunities: Contemporaneous and Long-Term Impacts of Rainfall Shocks on Human Capital," *Journal of Political Economy*, 2017, *125* (2), 527–561.

**Sims, Michelle J., Radost Stanimirova, Anton Raichuk, Maxim Neumann, Jessica Richter, Forrest Follett, James MacCarthy, Kristine Lister, Christopher Randle, Lindsey Sloat, Elena Esipova, Jaelah Jupiter, Charlotte Stanton, Dan Morris, Christy Melhart Slay, Drew Purves, and Nancy Harris**, "Global Drivers of Forest Loss at 1 km Resolution," *Environmental Research Letters*, 2025, *20* (7).

**Sonter, Laura J., Diego Herrera, Damian J. Barrett, Gillian L. Galford, Chris J. Moran, and Britaldo S. Soares-Filho**, "Mining Drives Extensive Deforestation in the Brazilian Amazon," *Nature Communications*, 2017, *8*, 1013.

**Steenhuisen, Frits and Simon J. Wilson**, "Development and Application of an Updated Geospatial Distribution Model for Gridding 2015 Global Mercury Emissions," *Atmospheric Environment*, 2019, *211*, 138–150.

**Sundberg, Ralph and Erik Melander**, "Introducing the UCDP Georeferenced Event Dataset," *Journal of Peace Research*, 2013, *50* (4), 523–532.

**Tang, Liang and Tim T. Werner**, "Global Mining Footprint Mapped from High-Resolution Satellite Imagery," *Communications Earth & Environment*, 2023, *4*, 134.

**Tollefsen, Andreas Forø, Håvard Strand, and Halvard Buhaug**, "PRIO-GRID: A Unified Spatial Data Structure," *Journal of Peace Research*, 2012, *49* (2), 363–374.

**von der Goltz, Jan and Prabhat Barnwal**, "Mines: The Local Wealth and Health Effects of Mineral Mining in Developing Countries," *Journal of Development Economics*, 2019, *139*, 1–16.

**World Bank**, "2020 State of the Artisanal and Small-Scale Mining Sector," Technical Report, World Bank and Pact, Delve, Washington, DC 2020.

# A Data

## A.1 Household Surveys

I assemble the Demographic and Health Surveys from the raw recode files of every standard survey in 33 African countries: 166 surveys fielded between 1986 and 2025 (Table A1). From each survey I keep the women's, men's, household, household-member and birth recodes, and I link each sampling cluster to its half-degree cell through the survey's GPS file. The 132 surveys with coordinates locate 65,746 clusters, and 99 percent of the people interviewed in them are linked to a cell. Ethiopian surveys record dates in the Ethiopian calendar, and I convert them to the Gregorian calendar by adding 92 months to the century-month codes, except in the 2024 survey, which already uses Gregorian dates.

Online Appendix OA1 compares this build with the survey file of Girard et al. (2025): of their 189 survey-years, 145 are in this build, and in those I match 97.6 percent of their women one to one.

# B Additional Results

This appendix reports the estimates behind Figures 3 and 5 and Section 5.2, the robustness checks of Section 6, which Figure 8 plots, and the other checks cited in Section 5.

Table A1—Demographic and Health Surveys by country

| Country | Survey years | Women | Men | Households | Fall | Second rise |
|---|---|---|---|---|---|---|
| Angola | 2015, 2023 | 29,096 | 12,288 | 32,352 | | Yes |
| Benin | 1996, 2001, (2006), 2012, 2017 | 43,731 | 16,792 | 41,376 | Yes | |
| Burkina Faso | 1993, 1999, 2003, 2010, 2021 | 58,909 | 22,670 | 45,803 | | |
| Burundi | (1987), 2010, 2016 | 26,591 | 11,815 | 24,517 | | |
| Cameroon | 1991, (1998), 2004, 2011, 2018 | 44,540 | 20,200 | 39,817 | | Yes |
| Chad | (1997), (2004), 2014 | 17,719 | 5,248 | 17,233 | Yes | |
| Congo Democratic Republic | 2007, 2013, 2023 | 54,755 | 25,305 | 51,706 | | |
| Cote d'Ivoire | 1994, 1998, 2012, 2021 | 35,816 | 16,028 | 32,217 | | |
| Egypt | (1988), 1992, 1995, 2000, 2003, 2005, 2008, 2014 | 104,735 | 0 | 119,674 | | |
| Eswatini | 2006 | 4,908 | 4,086 | 4,756 | | |
| Ethiopia | 2000, 2005, 2011, 2016, 2019, 2024 | 90,463 | 46,911 | 89,614 | | Yes |
| Gabon | (2000), 2012, 2019 | 19,408 | 12,477 | 21,419 | | |
| Ghana | (1988), 1993, 1998, 2003, 2008, 2014, 2022 | 44,200 | 23,693 | 59,269 | | |
| Guinea | 1999, 2005, 2012, 2018 | 34,608 | 13,012 | 26,302 | Yes | |
| Kenya | (1989), (1993), (1998), 2003, 2008, 2014, 2022, (2025) | 79,680 | 34,223 | 91,710 | | |
| Lesotho | 2004, 2009, 2014, 2023 | 27,297 | 12,072 | 36,599 | | |
| Liberia | 1986, 2007, 2013, 2019 | 29,390 | 14,213 | 24,967 | | |
| Madagascar | (1992), 1997, (2004), 2008, 2021 | 42,993 | 17,476 | 45,244 | | |
| Malawi | (1992), 2000, 2004, 2010, 2015, 2024 | 93,538 | 29,755 | 101,528 | | Yes |
| Mali | (1987), 1996, 2001, 2006, 2012, 2018, 2023 | 77,746 | 26,666 | 62,580 | Yes | Yes |
| Mozambique | (1997), (2003), 2011, 2022 | 26,910 | 9,410 | 28,149 | | |
| Namibia | (1992), 2000, 2006, 2013 | 26,409 | 11,285 | 25,277 | | |
| Niger | 1992, 1998, (2006), 2012 | 25,240 | 9,040 | 21,920 | | |
| Nigeria | 1990, 2003, 2008, 2013, 2018, 2024 | 169,021 | 60,489 | 168,706 | Yes | Yes |
| Rwanda | (1992), (2000), 2005, 2008, 2010, 2015, 2019, 2025 | 74,499 | 37,120 | 70,048 | Yes | Yes |
| Senegal | (1986), 1993, 1997, 2005, 2010, 2012, 2014, 2015, 2016, 2017, 2018, 2019, 2023 | 130,745 | 45,261 | 66,514 | Yes | Yes |
| Sierra Leone | 2008, 2013, 2019 | 38,990 | 17,487 | 32,816 | | |
| South Africa | (1998), 2016 | 8,514 | 3,618 | 11,083 | | |
| Tanzania | (1992), (1996), 1999, (2004), 2010, 2015, 2022 | 42,234 | 15,206 | 41,112 | | Yes |
| Togo | 1988, 1998, 2013 | 21,378 | 8,283 | 17,027 | | |
| Uganda | (1988), (1995), 2000, 2006, 2011, 2016 | 40,960 | 11,543 | 43,293 | | |
| Zambia | (1992), (1996), (2002), 2007, 2013, 2018, 2024 | 50,884 | 45,732 | 48,441 | Yes | Yes |
| Zimbabwe | (1988), (1994), 1999, 2005, 2010, 2015 | 33,389 | 25,258 | 35,366 | | |
| Total | 166 surveys, 132 with GPS | 1,649,296 | 664,662 | 1,578,435 | 8 | 10 |

*Notes*: Standard DHS surveys in Africa, 1986 to 2025, listed by the year of the survey's name. Years in parentheses are surveys without cluster coordinates, which cannot be linked to cells and do not enter the estimates. Women, men and households are interviews linked to a cell. Fall marks the eight countries with interviews both at the peak (2012–14) and after it (2015–18), by interview date, which can fall in the year after the survey's name; second rise marks the ten countries with interviews both in 2015–18 and in 2019–25.

Table B1—Work over the gold price cycle

| | Currently working (1) | Manual work (2) | Extraction (3) | Agriculture (4) | Sales (5) | Services (6) |
|---|---|---|---|---|---|---|
| *Panel A: All adults* | | | | | | |
| Suitable × 1990s | −2.25 | −0.91 | | −6.03 | 2.82 | 0.47 |
| | (2.29) | (1.19) | | (3.68) | (2.50) | (0.59) |
| Suitable × rise, 2006–11 | −8.41*** | −0.26 | 0.88 | −6.86** | 2.62* | −1.80*** |
| | (2.46) | (1.25) | (0.62) | (2.78) | (1.56) | (0.64) |
| Suitable × peak, 2012–14 | −7.50*** | 2.15* | 1.09 | −12.86*** | 4.51*** | −1.36* |
| | (2.46) | (1.19) | (0.70) | (2.79) | (1.57) | (0.77) |
| Suitable × after, 2015–18 | −6.25*** | −0.26 | 0.26 | −7.62*** | 1.72 | −1.49* |
| | (2.23) | (1.26) | (0.73) | (2.72) | (1.63) | (0.81) |
| After minus peak | 1.25 | −2.42*** | −0.83* | 5.24*** | −2.78*** | −0.13 |
| | (1.60) | (0.85) | (0.45) | (1.97) | (1.04) | (0.82) |
| Mean of outcome (%) | 63.6 | 10.1 | 0.9 | 29.6 | 19.7 | 4.2 |
| Observations | 637,014 | 620,509 | 408,237 | 620,509 | 620,509 | 620,509 |
| *Panel B: Men* | | | | | | |
| Suitable × 1990s | −3.73 | 3.80 | | −7.08 | −1.31 | −1.62 |
| | (3.89) | (2.40) | | (4.42) | (1.49) | (1.20) |
| Suitable × rise, 2006–11 | −9.30** | 2.49 | 1.32** | −9.66** | 1.41 | −1.83* |
| | (3.93) | (1.85) | (0.56) | (4.55) | (1.50) | (0.95) |
| Suitable × peak, 2012–14 | −4.56 | 5.58*** | 1.24** | −16.06*** | 2.18 | −0.65 |
| | (3.93) | (1.73) | (0.58) | (4.14) | (1.39) | (0.73) |
| Suitable × after, 2015–18 | −6.18* | 0.71 | 0.74 | −6.53 | 1.70 | −2.21** |
| | (3.61) | (1.59) | (0.75) | (4.15) | (1.53) | (0.86) |
| After minus peak | −1.62 | −4.87*** | −0.50 | 9.52*** | −0.48 | −1.56** |
| | (1.91) | (1.37) | (0.54) | (2.33) | (0.86) | (0.74) |
| Mean of outcome (%) | 79.0 | 19.0 | 1.3 | 40.8 | 10.0 | 4.6 |
| Observations | 181,812 | 185,152 | 133,282 | 185,152 | 185,152 | 185,152 |
| *Panel C: Women* | | | | | | |
| Suitable × 1990s | −1.15 | −2.96** | | −4.34 | 4.26 | 1.06* |
| | (2.60) | (1.35) | | (3.96) | (3.10) | (0.56) |
| Suitable × rise, 2006–11 | −7.52*** | −1.17 | 0.76 | −5.79* | 3.18 | −1.85*** |
| | (2.63) | (1.51) | (0.77) | (2.96) | (2.07) | (0.66) |
| Suitable × peak, 2012–14 | −8.14*** | 0.73 | 1.11 | −11.02*** | 5.50*** | −1.65* |
| | (2.55) | (1.37) | (0.89) | (3.03) | (2.09) | (0.96) |
| Suitable × after, 2015–18 | −6.05*** | −0.16 | 0.09 | −8.18*** | 1.46 | −1.06 |
| | (2.24) | (1.49) | (0.92) | (2.91) | (2.16) | (0.96) |
| After minus peak | 2.09 | −0.88 | −1.01* | 2.84 | −4.04*** | 0.59 |
| | (1.93) | (1.04) | (0.60) | (2.17) | (1.46) | (1.06) |
| Mean of outcome (%) | 57.4 | 6.3 | 0.7 | 24.8 | 23.8 | 4.0 |
| Observations | 455,201 | 435,356 | 274,954 | 435,356 | 435,356 | 435,356 |

*Notes*: Each column reports equation (2) for an indicator equal to 100 if the respondent's own occupation is in the sector and 0 otherwise, so that coefficients are in percentage points, in the eight countries surveyed both at the peak and after it (Benin, Chad, Guinea, Mali, Nigeria, Rwanda, Senegal and Zambia), surveys to 2018. Coefficients are the change in a fully gold-suitable cell relative to 2000 to 2005. Currently working (column 1) is defined as in Table 2. The window 2015–18 is reported both as its coefficient and as its difference from the peak. Panel A stacks the men's and women's interviews and controls for sex. Extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it; no survey before 2000 codes it, so column 3 has no 1990s window. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B2—Work in the second rise of the gold price, main specification

| | | Suitable × after, 2015–18 (1) | Suitable × second rise, 2019–25 (2) | Second rise minus after (3) | Mean (4) | Observations (5) |
|---|---|---|---|---|---|---|
| All adults | Manual work | 0.5 | 0.8 | 0.3 | 12.0 | 1,072,390 |
| | | (1.4) | (1.5) | (1.0) | | |
| | Agriculture | −7.3*** | −5.9** | 1.4 | 29.1 | 1,072,390 |
| | | (2.5) | (2.6) | (1.8) | | |
| | Sales | 2.1* | 1.5 | −0.7 | 15.9 | 1,072,390 |
| | | (1.2) | (1.3) | (1.0) | | |
| Men | Manual work | 0.3 | 0.4 | 0.1 | 21.4 | 323,711 |
| | | (1.7) | (2.0) | (1.7) | | |
| | Agriculture | −5.0 | −3.1 | 2.0 | 39.5 | 323,711 |
| | | (3.3) | (3.5) | (2.3) | | |
| Women | Currently working | −4.3 | −5.5** | −1.2 | 53.8 | 768,233 |
| | | (2.7) | (2.8) | (1.9) | | |
| | Sales | 2.2 | 1.8 | −0.4 | 18.8 | 748,678 |
| | | (1.4) | (1.5) | (1.3) | | |

*Notes*: Coefficients of equation (2) in a fully gold-suitable cell, relative to 2000–05, in percentage points, for each respondent's own occupation or current work, as an indicator equal to 100 or 0. Sample: the ten countries surveyed both in 2015–18 and in 2019–25 (Angola, Cameroon, Ethiopia, Malawi, Mali, Nigeria, Rwanda, Senegal, Tanzania and Zambia), all surveys to 2025. The regressions also include the 1990s, rise and peak windows, age and its square (and sex for all adults), and cell and country-by-year effects; the temperature and crop controls end in 2018 and are omitted. Column 3 is the difference between columns 2 and 1. Column 4 is the mean of the outcome. Data from the Demographic and Health Surveys. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B3—Persistent outcomes with a trend for gold-suitable cells

| | Attendance gap, ages 16–20 (1) | Wealth index (2) | Living-standards index (3) | ln(1 + night lights) (4) |
|---|---|---|---|---|
| *Panel A: Suitable × lagged price index, all years* | | | | |
| Without trend | −2.53* | 0.12** | 0.11** | 0.040*** |
| | (1.34) | (0.06) | (0.05) | (0.008) |
| With trend | −6.25** | 0.32*** | 0.22*** | 0.017*** |
| | (2.46) | (0.08) | (0.09) | (0.004) |
| Suitable × trend (per decade) | 2.29* | −0.14*** | −0.07 | 0.010*** |
| | (1.39) | (0.04) | (0.04) | (0.003) |
| Mean of outcome | −22.98 | 0.00 | 0.26 | 0.192 |
| Observations | 33,853 | 447,602 | 491,641 | 352,077 |
| *Panel B: Windows with the trend* | | | | |
| Suitable × 1990s | | | −0.08 | 0.012*** |
| | | | (0.12) | (0.002) |
| Suitable × peak, 2012–14 | −5.41 | 0.22** | 0.39*** | −0.003 |
| | (4.07) | (0.11) | (0.14) | (0.004) |
| After minus peak | 2.20 | −0.11 | 0.04 | 0.016*** |
| | (1.63) | (0.07) | (0.08) | (0.005) |
| Second rise minus after | −0.06 | −0.03 | 0.13 | 0.003 |
| | (2.56) | (0.08) | (0.08) | (0.003) |
| Suitable × trend (per decade) | 0.43 | 0.03 | −0.22* | 0.015*** |
| | (3.39) | (0.11) | (0.13) | (0.004) |
| Mean of outcome | −22.98 | 0.00 | 0.26 | 0.192 |
| Observations | 33,853 | 447,602 | 491,641 | 352,077 |

*Notes*: Column 1: the attendance gap between ages 16–20 and ages 6–11 in the same cell and survey, in percentage points, estimated as in Table B4 with the coefficients for ages 12–15 and 21–24 in the same regression (not shown), surveys from 2000, all countries. Columns 2 and 3: households in the eight countries surveyed at the peak and after it, surveys to 2025, in standard deviations. Column 4: all African cells, 1992–2024, with country-by-year effects for each brightness group before the boom, as in Table B4. Panel A replaces the windows with the suitable share times the lagged gold price index in each year (2013 equal to one); Panel B adds the trend to equation (2). The trend is the suitable share times decades since 2000 (for column 1, interacted with each age group). The Panel B regressions also include the rise (2006–11), after (2015–18) and second-rise (2019–25) windows, whose coefficients are not reported. Column 1 has no 1990s window, because its surveys start in 2000, and the wealth index of column 2 is not recorded in the 1990s surveys of these countries. Each panel reports the mean of the outcome and the observations of its regressions (in column 1, the mean gap over the three age groups, weighted as in the regression); the regression without the trend in Panel A uses the same observations. Cell (or cell-by-age-group) and country-by-year (or country-by-year-by-age-group) effects. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4—Human development over the gold price cycle

| | All countries | | | Eight countries surveyed at and after the peak | | | | |
|---|---|---|---|---|---|---|---|---|
| | Attendance 16–20 vs 6–11 (1) | Night lights ln(1 + DN) (2) | Under-five mortality (3) | Living-standards index (SD) (4) | Wealth index (SD) (5) | Electricity (6) | Improved water (7) | No toilet facility (8) |
| Suitable × price, eq. (1) | −2.5* | 0.040*** | 0.33 | 0.11** | 0.12** | 4.4* | 3.9 | −4.6** |
| | (1.3) | (0.008) | (0.46) | (0.05) | (0.06) | (2.3) | (2.8) | (2.2) |
| *Suitable × window, eq. (2)* | | | | | | | | |
| 1990s | | 0.002 | 0.20 | 0.05 | | 9.0*** | −9.8** | 3.5 |
| | | (0.002) | (0.34) | (0.09) | | (2.9) | (4.4) | (5.4) |
| Rise, 2006–11 | −2.0 | 0.008*** | −0.24 | 0.04 | 0.14 | 1.6 | 7.6* | 3.4 |
| | (1.4) | (0.002) | (0.27) | (0.08) | (0.09) | (3.1) | (4.6) | (3.0) |
| Peak, 2012–14 | −4.9*** | 0.013*** | 0.49 | 0.17*** | 0.25*** | 6.3** | 9.7** | −0.9 |
| | (1.6) | (0.004) | (0.39) | (0.06) | (0.09) | (2.7) | (4.0) | (2.4) |
| After, 2015–18 | −2.6 | 0.034*** | 0.47 | 0.10 | 0.15* | 7.8*** | 4.9 | 0.2 |
| | (1.7) | (0.007) | (0.43) | (0.06) | (0.08) | (2.7) | (4.0) | (2.5) |
| Second rise, 2019–25 | −2.4 | 0.045*** | 0.36 | 0.10 | 0.14 | 5.0 | 4.8 | −3.6 |
| | (1.7) | (0.009) | (1.00) | (0.07) | (0.08) | (3.1) | (4.2) | (3.0) |
| After minus peak | 2.3* | 0.022*** | −0.01 | −0.07 | −0.10** | 1.5 | −4.9** | 1.1 |
| | (1.4) | (0.005) | (0.33) | (0.05) | (0.04) | (2.3) | (2.3) | (1.7) |
| Second rise minus after | 0.2 | 0.011** | −0.11 | 0.00 | −0.02 | −2.8 | −0.1 | −3.9* |
| | (1.4) | (0.004) | (0.97) | (0.06) | (0.05) | (2.7) | (2.8) | (2.1) |
| *Holm-adjusted p, columns 1 to 4* | | | | | | | | |
| Suitable × price | [0.12] | [<0.01] | [0.47] | [0.10] | | | | |
| Peak | [0.01] | [<0.01] | [0.21] | [0.02] | | | | |
| After minus peak | [0.28] | [<0.01] | [0.96] | [0.32] | | | | |
| Second rise minus after | [1.00] | [0.04] | [1.00] | [1.00] | | | | |
| Mean of outcome | −23.0 | 0.192 | 11.30 | 0.26 | 0.00 | 38.1 | 65.2 | 25.5 |
| Observations | 33,853 | 352,077 | 1,948,162 | 491,641 | 447,602 | 491,661 | 490,049 | 490,522 |
| Countries | 33 | 54 | 33 | 8 | 8 | 8 | 8 | 8 |

*Notes*: The first row reports equation (1) and the rows below it equation (2); column 1 uses equation (3). Column 1 is the attendance rate of household members aged 16 to 20 minus that of members aged 6 to 11 in the same cell and survey, in percentage points, in all countries with household rosters recording attendance, from 2000; observations are cell-survey-age-group means, weighted by the number of household members in the age group, with cell-by-age-group and country-by-year-by-age-group effects; its mean is the gap averaged over the three age groups in the regression (12–15, 16–20 and 21–24), weighted as in the regression. Column 2 is the logarithm of one plus mean harmonised night lights in all African cells, 1992 to 2024, with country-by-year effects for each brightness group before the boom: unlit in 1992–2005, or a tercile of mean lights then among lit cells. Column 3 is an indicator equal to 100 if a child died before age five, for births in the 15 years before interview and at least 60 months before it, by year of birth, with cell-by-survey and country-by-birth-year effects and controls for the mother's age at birth and its square, sex and twin birth. Columns 4 to 8 use households in the eight countries surveyed both at the peak and after it: the wealth index, standardised within each survey (column 5), and indicators equal to 100 or 0 for electricity, an improved source of drinking water (piped, borehole, protected well or spring, rainwater or bottled) and no toilet facility (columns 6 to 8). The living-standards index (column 4) is the mean of the four, each standardised by its mean and standard deviation among households surveyed in 2000 to 2005 and no toilet facility entering with a negative sign, for households with at least three of them, divided by its standard deviation in 2000 to 2005 (Kling et al., 2007; Anderson, 2008). Columns 4 to 8 have cell and country-by-year effects. Attendance (column 1) starts in 2000 and the wealth index (column 5) is not recorded in the 1990s surveys of these countries, so neither has a 1990s window. Holm $p$-values adjust for testing four outcomes, columns 1 to 4, separately for each row (Holm, 1979). Surveys to 2025 and night lights to 2024, without the temperature and crop controls. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B5—Human development by age and group

| | Rise, 2006–11 (1) | Peak, 2012–14 (2) | After, 2015–18 (3) | Second rise, 2019–25 (4) | Mean of outcome (5) | Observations (6) |
|---|---|---|---|---|---|---|
| *Panel A: Attendance gap against ages 6–11, by age group (percentage points)* | | | | | | |
| Ages 12–15 | −2.4** | −3.0*** | −1.2 | −2.1* | −23.0 | 33,853 |
| | (1.1) | (1.1) | (1.3) | (1.1) | | |
| Ages 16–20 | −2.0 | −4.9*** | −2.6 | −2.4 | −23.0 | 33,853 |
| | (1.4) | (1.6) | (1.7) | (1.7) | | |
| Ages 21–24 | −1.3 | 1.0 | 2.0 | 2.4 | −23.0 | 33,853 |
| | (1.8) | (1.9) | (1.8) | (1.8) | | |
| *Panel B: Attendance gap at ages 16–20, by group (percentage points)* | | | | | | |
| Boys | −1.1 | −4.6** | −2.3 | −2.5 | −18.7 | 33,093 |
| | (1.8) | (1.9) | (1.9) | (2.0) | | |
| Girls | −3.1* | −5.1*** | −2.8 | −1.8 | −27.0 | 33,385 |
| | (1.7) | (1.9) | (1.9) | (2.0) | | |
| Head with no schooling | −2.2 | −6.2*** | −3.1 | −6.1*** | −15.7 | 28,975 |
| | (2.1) | (2.1) | (2.4) | (2.2) | | |
| Head with some schooling | −1.6 | −4.3** | −2.4 | −0.7 | −27.7 | 31,491 |
| | (1.8) | (2.0) | (2.0) | (2.0) | | |
| *Panel C: Wealth index, eight countries (standard deviations)* | | | | | | |
| Rural households | 0.07 | 0.17** | 0.13* | 0.11 | −0.44 | 284,671 |
| | (0.08) | (0.08) | (0.08) | (0.08) | | |
| Urban households | 0.05 | 0.07 | −0.05 | −0.01 | 0.78 | 162,931 |
| | (0.12) | (0.14) | (0.15) | (0.16) | | |
| *Panel D: Night lights, ln(1 + DN)* | | | | | | |
| Cells dark in 1992–2005 | 0.001* | 0.003 | 0.031*** | 0.030*** | 0.040 | 136,925 |
| | (0.000) | (0.003) | (0.009) | (0.011) | | |
| Cells lit in 1992–2005 | 0.015*** | 0.022*** | 0.038*** | 0.061*** | 0.363 | 171,336 |
| | (0.004) | (0.007) | (0.012) | (0.014) | | |
| *Panel E: Under-five mortality (deaths per 100 births)* | | | | | | |
| Mother with no schooling | 0.0 | 0.4 | −0.1 | 0.8 | 13.8 | 949,093 |
| | (0.4) | (0.6) | (0.7) | (1.8) | | |
| Mother with some schooling | −0.4 | 0.6 | 0.8 | 0.0 | 8.9 | 998,501 |
| | (0.4) | (0.5) | (0.5) | (1.2) | | |
| Boys | −0.1 | 0.8 | 0.5 | 1.2 | 12.0 | 990,743 |
| | (0.4) | (0.6) | (0.6) | (1.5) | | |
| Girls | −0.4 | 0.4 | 0.6 | −0.4 | 10.6 | 957,328 |
| | (0.4) | (0.5) | (0.6) | (1.4) | | |
| Rural mother | −0.3 | 0.5 | 0.2 | 0.2 | 12.5 | 1,394,049 |
| | (0.3) | (0.5) | (0.5) | (1.2) | | |
| Urban mother | −0.1 | 0.5 | 1.1* | 0.8 | 8.4 | 554,102 |
| | (0.4) | (0.6) | (0.7) | (1.8) | | |

*Notes*: Coefficients of equation (2) in a fully gold-suitable cell, relative to 2000–05, with the specifications of Appendix Table B4. The three rows of Panel A come from one regression; every other row is a separate regression. Columns 5 and 6 report the mean of the outcome and the observations of each regression. Panel A plots in Figure 6 and Panels B to E in Figure 7. Panels A and B: equation (3), all countries, from 2000; observations are cell-survey-age-group means, and the mean is the gap to ages 6–11 averaged over the three age groups in the regression (12–15, 16–20 and 21–24), weighted as in the regression. Panel C: households in the eight countries surveyed at the peak and after it. Panel D: all African cells, with cells dark or lit by mean night lights in 1992–2005, the second rise ending in 2024. Panel E: births by year of birth, with the mother's schooling and residence at interview. The table omits the 1990s window: attendance (Panels A and B) starts in 2000, the wealth index (Panel C) is not recorded in the 1990s surveys of these countries, and cells dark in 1992–2005 (Panel D) are estimated from 2000; the regressions for lit cells (Panel D) and for mortality (Panel E) include a 1990s window, which is not reported. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6—Robustness of the cycle estimates

| | Men | | Women | |
|---|---|---|---|---|
| | Manual work (1) | Agriculture (2) | Sales (3) | Currently working (4) |
| *Panel A: Suitable × peak, 2012–14* | | | | |
| Main | 5.58*** (1.73) | −16.06*** (4.14) | 5.50*** (2.09) | −8.14*** (2.55) |
| Any suitable land in the cell | 4.54*** (1.27) | −9.65*** (2.31) | 6.96*** (1.35) | −6.66*** (1.68) |
| Suitable share of at least 0.5 | 2.29* (1.29) | −8.78** (3.42) | 2.16 (1.43) | −5.06** (2.26) |
| Share within 10 km of suitable land | 4.82*** (1.50) | −12.50*** (3.01) | 6.12*** (1.71) | −7.13*** (2.09) |
| Industrial gold mines by 2000 × price | 5.51*** (1.73) | −15.74*** (4.16) | 5.26** (2.10) | −7.70*** (2.56) |
| All industrial gold mines × price | 5.75*** (1.75) | −15.64*** (4.27) | 4.79** (2.12) | −7.52*** (2.65) |
| Other industrial mines × own price | 5.99*** (1.68) | −15.13*** (4.11) | 5.24** (2.11) | −7.79*** (2.55) |
| Conflict events in the cell-year | 5.58*** (1.72) | −16.05*** (4.13) | 5.47*** (2.08) | −8.14*** (2.55) |
| No temperature and crop controls | 5.55*** (1.73) | −15.83*** (4.14) | 5.32*** (2.05) | −8.17*** (2.54) |
| Reference period 1990–2005 | 4.45*** (1.71) | −13.94*** (3.81) | 3.99* (2.40) | −7.73*** (2.38) |
| Region-by-year effects | 5.65*** (1.99) | −8.60** (3.54) | 3.98* (2.13) | −3.73 (2.53) |
| Rural clusters only | 3.65** (1.80) | −13.99*** (5.00) | 4.94** (2.43) | −12.55*** (3.24) |
| Urban residence × window | 4.44*** (1.71) | −13.57*** (4.00) | 4.85** (2.03) | −8.13*** (2.56) |
| Cells observed in every window | 5.28*** (1.90) | −15.41*** (4.28) | 5.74*** (2.20) | −9.13*** (2.74) |
| Non-migrants only | 8.15*** (2.92) | −16.02*** (5.75) | 5.03 (3.69) | −2.22 (4.81) |
| No clusters near a cell edge | 5.79*** (1.85) | −14.45*** (4.39) | 5.78** (2.42) | −7.68*** (2.72) |
| Other mines, conflict and urban × window | 4.87*** (1.67) | −12.75*** (3.87) | 4.57** (2.05) | −7.81*** (2.56) |
| Drop one country, range | [4.8, 7.3] | [−19.1, −12.3] | [3.3, 6.8] | [−11.8, −5.2] |
| Country jackknife $p$ | [0.04] | [0.02] | [0.07] | [0.14] |
| Wild bootstrap $p$, countries | [0.12] | [0.02] | [0.08] | [0.16] |
| Placebo permutation $p$ | [0.005] | [<0.001] | [0.009] | [0.001] |
| *Panel B: After minus peak* | | | | |
| Main | −4.87*** (1.37) | 9.52*** (2.33) | −4.04*** (1.46) | 2.09 (1.93) |
| Any suitable land in the cell | −3.50*** (1.09) | 6.97*** (1.71) | −3.60*** (1.23) | 0.83 (1.41) |
| Suitable share of at least 0.5 | −2.84** (1.14) | 5.31*** (1.85) | −2.32** (1.12) | 2.26 (1.55) |
| Share within 10 km of suitable land | −4.26*** (1.22) | 8.64*** (1.96) | −3.75*** (1.35) | 1.57 (1.64) |
| Industrial gold mines by 2000 × price | −4.85*** (1.37) | 9.46*** (2.33) | −4.00*** (1.46) | 2.02 (1.93) |
| All industrial gold mines × price | −4.91*** (1.37) | 9.43*** (2.34) | −3.88*** (1.47) | 1.92 (1.94) |
| Other industrial mines × own price | −4.91*** (1.37) | 9.44*** (2.35) | −4.01*** (1.47) | 2.05 (1.93) |
| Conflict events in the cell-year | −4.84*** (1.37) | 9.41*** (2.31) | −3.94*** (1.47) | 2.12 (1.93) |
| No temperature and crop controls | −4.84*** (1.37) | 9.29*** (2.30) | −4.04*** (1.48) | 2.08 (1.93) |
| Reference period 1990–2005 | −4.89*** (1.37) | 9.56*** (2.33) | −4.05*** (1.46) | 2.09 (1.93) |
| Region-by-year effects | −1.14 (1.62) | 7.19** (2.98) | 1.06 (2.11) | 2.44 (2.08) |
| Rural clusters only | −6.90*** (1.84) | 12.03*** (2.94) | −4.94*** (1.80) | 4.45* (2.49) |
| Urban residence × window | −4.29*** (1.36) | 7.94*** (2.15) | −3.83*** (1.44) | 2.01 (1.91) |
| Cells observed in every window | −3.41* (1.74) | 7.57*** (2.66) | −3.88** (1.72) | 3.49 (2.29) |
| Non-migrants only | −5.75** (2.46) | 13.00*** (4.79) | −2.88 (2.73) | −0.47 (4.27) |
| No clusters near a cell edge | −3.10** (1.54) | 6.19** (2.54) | −4.71*** (1.69) | 0.21 (2.00) |
| Other mines, conflict and urban × window | −4.32*** (1.35) | 7.79*** (2.15) | −3.73*** (1.45) | 2.00 (1.91) |
| Drop one country, range | [−5.7, −3.2] | [7.9, 10.6] | [−5.4, −2.2] | [0.3, 5.8] |
| Country jackknife $p$ | [0.05] | [0.01] | [0.17] | [0.63] |
| Wild bootstrap $p$, countries | [0.18] | [0.16] | [0.30] | [0.51] |
| Placebo permutation $p$ | [0.005] | [<0.001] | [0.019] | [0.384] |
| *Panel C: Suitable × lagged price index, all years* | | | | |
| Without trend | 2.89* (1.59) | −11.77*** (3.19) | 4.03* (2.35) | −7.03*** (2.21) |
| With suitable × linear trend | 8.66*** (2.38) | −21.85*** (4.27) | 10.22*** (2.57) | −6.62* (3.48) |
| Suitable × trend (per decade) | −4.39*** (1.37) | 7.66*** (2.35) | −4.38** (1.74) | −0.29 (1.82) |

*Notes*: Panels A and B report equation (2) for the respondent's own occupation or current work, in percentage points, in the eight countries surveyed both at the peak and after it, surveys to 2018, with the change named in the row; Figure 8 plots them. The first row is the specification of Appendix Table B1: temperature, the crop-price index, age and age squared, with cell and country-by-year effects. Any suitable land and a share of at least 0.5 are indicators. Industrial gold mines are cells with a mine producing by 2000, or at any date, times the lagged price; other industrial mines are the price index of Girard et al. (2025), in which each industrial mine of any mineral is valued at the world price of its main mineral. Conflict events are $\ln(1+\text{events})$ in the cell and interview year (Sundberg and Melander, 2013). With the 1990–2005 reference, the 1990s window is dropped. Region-by-year effects use the survey regions. Urban residence × window adds an urban indicator and its interaction with each window; cells observed in every window have respondents in 2000–05, at the peak and after it. Non-migrants have always lived in the place or for at least five years. Clusters near a cell edge lie within the DHS displacement distance of it. Dropping one country reports the smallest and largest of the eight estimates that each leave out one country. The country jackknife (CV3) uses those eight estimates with $t(7)$ critical values. Bootstrap $p$-values are from the wild cluster restricted bootstrap with Rademacher weights over all 256 draws. Placebo $p$-values are the share of 1,000 reassignments of the suitable share across cells within country, on cell-by-year means, with a larger absolute estimate. Panel C replaces the windows with the suitable share times the lagged gold price index (2013 equal to one) in each year; the trend is the suitable share times decades since 2000. Observations (men's outcomes; women's sales; women's current work): main sample 185,152; 435,356; 455,201, in 1,166 cells; without temperature and crop controls 185,161; 442,825; 462,685; with region-by-year effects 185,149; 435,355; 455,200; rural clusters 115,541; 277,169; 288,993; cells observed in every window 131,575; 337,089; 352,713; non-migrants 85,620; 187,859; 199,414; without clusters near a cell edge 138,917; 324,959; 339,299. All other rows use the main sample. Standard errors clustered by cell in parentheses; stars refer to them. Brackets give the range of estimates in the drop-one-country rows and $p$-values in the jackknife, bootstrap and placebo rows. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table B7—The cycle in all countries, and the second rise by cell

| | Men | | Women | |
|---|---|---|---|---|
| | Manual work (1) | Agriculture (2) | Sales (3) | Currently working (4) |
| *Panel A: All countries* | | | | |
| Suitable × peak, 2012–14 | 2.65** (1.28) | −6.39*** (2.12) | 4.13*** (1.24) | −2.43 (1.90) |
| After minus peak | −3.11*** (1.15) | 6.12*** (1.92) | −3.13*** (1.18) | 1.07 (1.68) |
| Second rise minus after, all surveys | 1.83 (1.33) | −1.63 (1.87) | 1.56 (1.10) | −3.69** (1.64) |
| Mean of outcome, surveys to 2018 | 18.4 | 41.2 | 16.5 | 54.7 |
| Mean of outcome, all surveys | 20.0 | 39.1 | 16.2 | 54.4 |
| Observations, surveys to 2018 | 475,389 | 475,389 | 1,182,257 | 1,217,450 |
| Observations, all surveys | 642,301 | 642,301 | 1,550,325 | 1,587,147 |
| Countries | 32 | 32 | 33 | 33 |
| *Panel B: Second rise minus after, ten rise countries, by cell* | | | | |
| Conflict events in 2019–24 | 1.76 (2.61) | 2.08 (4.00) | −3.35 (2.69) | |
| Mean of outcome | 19.8 | 38.2 | 22.2 | |
| Observations | 127,863 | 127,863 | 298,376 | |
| Countries | 10 | 10 | 10 | |
| No conflict events in 2019–24 | −0.90 (2.28) | 2.80 (2.89) | 1.72 (1.37) | |
| Mean of outcome | 22.4 | 40.3 | 16.6 | |
| Observations | 195,848 | 195,848 | 450,302 | |
| Countries | 10 | 10 | 10 | |
| Mapped mine in the cell | 9.08 (7.75) | −12.10 (7.64) | 1.64 (3.24) | |
| Mean of outcome | 27.9 | 30.4 | 14.6 | |
| Observations | 20,557 | 20,557 | 42,257 | |
| Countries | 8 | 8 | 8 | |
| No mapped mine | 0.12 (1.69) | 2.32 (2.29) | −0.22 (1.35) | |
| Mean of outcome | 21.0 | 40.1 | 19.0 | |
| Observations | 303,154 | 303,154 | 706,421 | |
| Countries | 10 | 10 | 10 | |

*Notes*: Each entry reports equation (2) for the respondent's own occupation or current work, in percentage points, relative to 2000–05. Panel A uses every country: the first two rows use surveys to 2018 with the controls of Appendix Table B1; the third row uses all surveys with age controls, as in Online Appendix Table OA2. Panel B repeats Appendix Table B2, column 3, in the ten countries surveyed both in 2015–18 and in 2019–25, for cells with and without a UCDP conflict event in 2019–24 (Sundberg and Melander, 2013) and with and without a mine on the satellite map of Tang and Werner (2023); cells with a mapped mine hold 6 percent of respondents and lie in eight of the ten countries. Panel B covers the three occupations that moved at the peak and returned after it; women's current work did not change significantly after the peak (Appendix Table B1), so column 4 is blank in Panel B. Means of the outcome and observations refer to the regressions in each row; in Panel A, the first two rows use the surveys to 2018 and the third all surveys. All regressions have cell and country-by-year effects. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B8—Further estimates and tests cited in the text: men's manual work and agriculture, women's sales and current work

| | Men | | Women | |
|---|---|---|---|---|
| | Manual work | Agriculture | Sales | Currently working |
| *Panel A: Change per unit of the price index at the interview dates, eight countries* | | | | |
| Boom: peak relative to 2000–05 | 6.4 | −18.5 | 6.3 | −9.3 |
| Fall: peak minus after | 20.8 | −40.7 | 17.2 | −8.9 |
| *p*, fall equals boom | [0.01] | [0.02] | [0.07] | [0.95] |
| *Panel B: Second rise minus after, ten countries* | | | | |
| Estimate | 0.1 | 2.0 | −0.4 | −1.2 |
| Implied by the response during the boom | 3.1 | −7.0 | 2.7 | −3.0 |
| *p*, estimate equals it | [0.12] | [<0.01] | [0.03] | [0.47] |
| *p*, with the real price | [0.21] | [<0.01] | [0.06] | [0.62] |
| Implied by the response after the peak | 10.2 | −15.8 | 5.7 | −3.0 |
| *p*, estimate equals it | [<0.01] | [<0.01] | [0.01] | [0.56] |
| *Panel C: p-values for the peak (first) and for after minus peak (second), eight countries* | | | | |
| Conley *p*, 100 km | [<0.01] [0.01] | [<0.01] [<0.01] | [0.03] [0.05] | [0.03] [0.40] |
| Conley *p*, 500 km | [<0.01] [0.01] | [<0.01] [<0.01] | [<0.01] [0.13] | [0.01] [0.37] |
| Shifted-map placebo *p* | [0.13] [0.17] | [0.05] [0.07] | [0.18] [0.32] | [0.25] [0.68] |
| Holm *p* across 24 cycle statistics | [0.02] [0.01] | [<0.01] [<0.01] | [0.10] [0.08] | [0.03] [1.00] |
| *Panel D: Five countries in both comparisons (Mali, Nigeria, Rwanda, Senegal, Zambia)* | | | | |
| Peak | 4.0 (2.4) | −19.5 (6.3) | 7.6 (2.3) | |
| After minus peak | −5.8 (1.7) | 9.2 (2.8) | −4.0 (1.8) | |
| Second rise minus after | −1.0 (1.9) | 3.6 (2.8) | 0.3 (1.9) | |
| Implied by the response after the peak | 10.6 | −15.9 | 6.9 | |
| *Panel E: Fieldwork timing and the local-currency real price, eight countries* | | | | |
| Windows by survey: peak / after minus peak | 6.0 / −5.7 | −16.5 / 11.0 | 5.5 / −4.2 | −6.4 / 0.4 |
| Country-year-month effects: peak / after minus peak | 4.6 / −2.5 | −12.5 / 8.1 | 5.6 / −2.7 | −7.0 / 4.1 |
| Local real price: boom / fall per unit | 7.7 / 29.9 | −22.2 / −58.5 | 7.5 / 23.9 | −11.1 / −12.4 |
| *p*, fall equals boom | [<0.01] | [0.01] | [0.05] | [0.90] |
| Second rise: implied by the boom in local prices | 2.4 | −5.6 | 2.2 | −2.4 |
| *p*, estimate equals it | [0.21] | [<0.01] | [0.06] | [0.62] |
| Ten countries: suitable × price / × price after 2019 | 9.3 / −1.4 | −17.2 / 5.8 | 7.2 / −2.6 | −4.3 / 2.3 |
| *Panel F: Men's manual work in the seven countries whose surveys code extraction* | | | | |
| Peak / after minus peak | 6.9 (2.0) / −5.3 (1.4) | | | |
| *Panel G: Other statistics cited in the text* | | | | |
| Forest loss per unit of the index, ln(1 + km$^2$) | 0.26 (0.03); with trend 0.17 (0.03); with trend and cover-tercile effects 0.12 (0.04) | | | |
| Urban share of surveyed households (%), eight countries | peak 8.6 (5.1); after minus peak −5.0 (2.3); mean 36 | | | |
| Men aged 15–19 in agriculture (%), eight countries | peak −17.7 (5.3); after minus peak 11.9 (3.6); mean 32 | | | |
| Conflict events, ln(1 + events), eight countries | peak −0.053 (0.026); after minus peak −0.061 (0.021) | | | |
| Night lights, DMSP years 1992–2013 only | peak 0.012 (0.003) | | | |
| Mapped mine footprint in the cell (%), no industrial gold mine | suitable share 5.4 (1.8); mean 2.7 | | | |
| Completed schooling (years), ages 21–45 | teen price −0.14 (0.11); with cohort trend 0.07 (0.12); implied by attendance −0.19; detectable with 80% power 0.32 | | | |
| Correlation of the lagged price index with the year, 1990–2018 | 0.83 | | | |

*Notes*: Percentage points in a fully gold-suitable cell unless stated. Panel A divides the peak coefficient of Appendix Table B1 by the change in the mean lagged price index at the interview dates from 2000–05 to the peak, and the change after the peak by the fall in that index from the peak to 2015–18; both rows are signed in the direction of the price, so a negative value means the outcome falls as the price rises and rises as it falls; the *p*-value tests the difference with cell-clustered standard errors. Panel B compares the estimate of Online Appendix Table OA2 with the change implied by scaling each response by the rise in the index from 2015–18 to 2019–25, in nominal dollars and deflated by US consumer prices. Each cell of Panel C holds two *p*-values: the first is for the peak coefficient and the second for after minus peak. The rows use Conley standard errors with a uniform kernel; the share of about 320 shifts of the suitability map by at least two degrees that give an estimate as large as the actual one; and Holm-adjusted *p*-values across the 24 cycle statistics of Appendix Table B1: the peak coefficient and the change after the peak for manual work, agriculture, sales and current work, for all adults, men and women. Panel D uses the specification of Appendix Table B1 for the cycle (surveys to 2018) and that of Online Appendix Table OA2 for the second rise (surveys to 2025, age controls only), in the five countries, and its implied change scales the change from 2015–18 to the peak by the change in the mean lagged price index at the interview dates; Panel F uses the specification of Appendix Table B1 in the seven countries. In Panel E, windows by survey assign each respondent to the window of the first interview year of the survey; country-year-month effects replace the country-by-year effects and add the suitable share times calendar-month indicators; the local real price is the dollar gold price times the official exchange rate, divided by the consumer price index (World Bank), indexed to 2013 within each country and lagged one year; the last row uses the ten rise countries, all surveys, with the suitable share times a linear trend. Panel G: estimates with standard errors clustered by cell in parentheses, from equation (2) or (1) in the samples named; forest loss in all forested cells with the trend in decades since 2000; the mine footprint from the map of Tang and Werner (2023), a cross-section with country effects and country clusters; completed schooling on the suitable share times the mean lagged price at ages 12 to 20, with cell-by-survey and country-by-birth-year effects. Panels D and F report standard errors clustered by cell in parentheses, except for the implied change in Panel D, which has none; Panels A, B, C and E report point estimates and *p*-values only. All *p*-values are in brackets.

Table B9—Who is surveyed over the gold price cycle

| | 1990s (1) | Rise 2006–11 (2) | Peak 2012–14 (3) | After minus peak (4) | Mean (5) | Obser-vations (6) |
|---|---|---|---|---|---|---|
| Household size | 1.39*** | 0.63** | 0.40 | −0.04 | 5.85 | 387,424 |
| | (0.35) | (0.28) | (0.27) | (0.10) | | |
| Men 15–29 (% of members) | −0.2 | 0.9 | 1.0 | 0.4 | 12.6 | 387,424 |
| | (0.9) | (0.7) | (0.7) | (0.5) | | |
| Female (% of members) | 0.5 | −2.0** | −0.8 | −0.1 | 50.1 | 387,424 |
| | (0.8) | (0.8) | (0.7) | (0.5) | | |
| Head with no schooling (%) | −2.1 | −1.3 | −3.3* | 1.8 | 47.0 | 387,410 |
| | (2.7) | (2.2) | (1.9) | (1.3) | | |
| Women arrived <5 years ago (%) | 5.7*** | −0.1 | −1.5 | 6.5** | 19.5 | 256,459 |
| | (2.2) | (2.8) | (3.7) | (3.0) | | |

*Notes*: Each row reports equation (2) in the eight countries surveyed both at the peak and after it, surveys to 2018 (so there is no 2019–25 window), with cell and country-by-year effects and no controls. The first four rows use households from the household rosters; the last uses women aged 15 to 49, with an indicator equal to 100 if the woman has lived in the place for less than five years. Coefficients are the change in a fully gold-suitable cell relative to 2000 to 2005. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B10—Child mortality in towns that predate the boom

| | Lit at night in 1992–99 | | | | Urban cluster before 2006 | | | |
|---|---|---|---|---|---|---|---|---|
| | Towns | | Other cells | | Towns | | Other cells | |
| | Under five (1) | Infant (2) | Under five (3) | Infant (4) | Under five (5) | Infant (6) | Under five (7) | Infant (8) |
| Rise, 2006–11 | −0.05 | −0.27 | −0.75 | −0.74 | 0.25 | 0.06 | −0.52 | −0.69** |
| | (0.30) | (0.26) | (0.62) | (0.56) | (0.38) | (0.32) | (0.37) | (0.33) |
| Peak, 2012–14 | 0.70 | 0.32 | −0.45 | −0.39 | 1.03** | 0.58 | −0.03 | −0.18 |
| | (0.43) | (0.37) | (0.91) | (0.77) | (0.52) | (0.44) | (0.56) | (0.47) |
| After, 2015–18 | 1.01** | 0.64 | −1.07 | −0.25 | 1.20** | 0.78 | 0.01 | 0.15 |
| | (0.46) | (0.41) | (1.04) | (0.85) | (0.61) | (0.55) | (0.60) | (0.50) |
| After minus peak | 0.31 | 0.31 | −0.62 | 0.15 | 0.18 | 0.20 | 0.04 | 0.34 |
| | (0.36) | (0.31) | (0.79) | (0.59) | (0.48) | (0.44) | (0.45) | (0.35) |
| Mean | 11.0 | 6.7 | 12.9 | 7.7 | 10.5 | 6.5 | 12.1 | 7.3 |
| Births | 1,608,991 | 1,608,991 | 339,165 | 339,165 | 1,013,257 | 1,013,257 | 934,899 | 934,899 |
| Cells | 2,573 | 2,573 | 1,899 | 1,899 | 858 | 858 | 3,614 | 3,614 |

*Notes*: Each column reports equation (2), suitable × window, for an indicator equal to 100 if the child died before age five (under five) or age one (infant), by year of birth, for births in the 15 years before interview and at least 60 months before it, in all countries, with cell-by-survey and country-by-birth-year effects and controls for the mother's age at birth and its square, the child's sex and twin birth. Columns 1 to 4 define towns as cells with positive mean harmonised night lights in 1992 to 1999; columns 5 to 8 as cells with at least one urban DHS cluster in a survey fielded before 2006. The mean is deaths per 100 births. The regressions also include the 1990s and 2019–25 windows, which are not reported; because births must have occurred at least 60 months before an interview in 2025 or earlier, the 2019–25 window holds only births of 2019 and 2020. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

# Online Appendix

## OA1 Validation of the Survey Build

The survey file of Girard et al. (2025) covers 189 survey-years. Of these, 145 are in this build; the other 44, which hold 15 percent of the women in their file, are mostly malaria and AIDS indicator surveys, which this build excludes. In the shared survey-years, I match 97.6 percent of their women one to one on survey, year and month of interview, age and cell, and across surveys the number of women per cell correlates at a median of 0.994.

In Table OA1, I re-estimate their specification for the measure that counts the woman's or her partner's occupation. On this build and their survey-years, manual work rises by 3.8 points per unit of the price index, against 2.8 on their file, extraction by 1.1 on both, and agriculture falls by 2.5 against 2.8 (rows 1 and 2). Services differ, at 0.8 against 3.0 points. This build holds more women in the same survey-years, 1,154,443 against 869,054, and keeping one woman per household brings the count to 824,370 and the estimate for services to 1.4 points (row 3). Adding the surveys to 2018 that their file does not contain leaves the estimates close to row 2 (row 4).

Table OA1—Reproducing Girard et al. (2025) on the new survey build

| | Manual (1) | Extraction (2) | Agriculture (3) | Services (4) | Women |
|---|---|---|---|---|---|
| Girard et al. file | 2.84** | 1.06 | −2.77 | 3.02** | 869,054 |
| | (1.18) | (0.73) | (1.71) | (1.39) | |
| Same survey-years | 3.82*** | 1.09** | −2.49 | 0.76 | 1,154,443 |
| | (1.12) | (0.44) | (1.84) | (1.50) | |
| One woman per household | 4.12*** | 1.16** | −2.55 | 1.42 | 824,370 |
| | (1.24) | (0.47) | (1.85) | (1.56) | |
| All surveys to 2018 | 3.88*** | 1.08** | −2.48 | 1.18 | 1,177,619 |
| | (1.09) | (0.43) | (1.83) | (1.49) | |

*Notes*: Each entry is the coefficient on the suitable share times the lagged gold price index (2013 equal to one) for an indicator equal to 100 if the woman or her partner works in the sector, the outcome of Girard et al. (2025), in their specification: cell and country-by-year effects, temperature, crop suitability times crop prices, industrial gold times price, age and age squared, surveys to 2018. Sectors follow the DHS occupation groups of the woman's (v717) and her partner's (v705) occupation; extraction is manual work whose occupation label names mining or quarrying, in the surveys whose labels code it. The first row uses their survey file; rows 2 to 4 use the survey files assembled for this paper. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## OA2 The Second Rise against the First Boom

Section 5.2 reports how work changed between 2015–18 and 2019–25 in the ten countries surveyed in both windows. This appendix compares that change with what the first boom's response implies, splits it by earlier mining and by age, and estimates it with

the yearly price.

Two benchmarks scale the earlier responses to the second rise. The first scales the response during the boom in the ten countries, the change from 2000–05 to the peak (when the mean lagged index at the interview dates rose from 0.33 to 1.13), by the rise in the lagged index from 2015–18 to 2019–25 relative to its rise from 2000–05 to the peak. On that scale, men's farming in a fully gold-suitable cell should then have fallen by 7.0 points and their manual work risen by 3.1. The second scales the change from the peak to 2015–18, estimated in the same regression and the same ten countries, by $k = (\bar{P}_{19\text{–}25} - \bar{P}_{15\text{–}18})/(\bar{P}_{12\text{–}14} - \bar{P}_{15\text{–}18}) = 1.70$, the ratio of the rise in the mean lagged index at the interview dates in the second rise (from 0.88 to 1.31) to its fall after the peak (from 1.13 to 0.88). On that scale, farming should have fallen by 15.8 points and manual work risen by 10.2. The response after the peak is steeper per unit of the price, so the second benchmark is the more demanding one. In the ten countries, the peak coefficient is identified by those also surveyed in 2012–14.

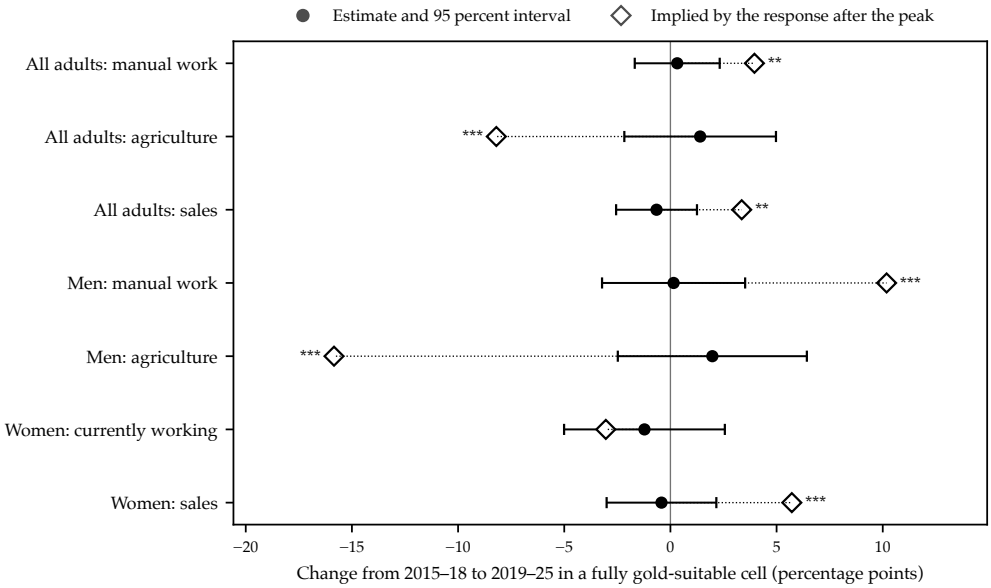



Figure OA1—Work in the second rise of the gold price

*Notes*: Each row plots the difference between the 2019–25 and 2015–18 coefficients of equation (2) in a fully gold-suitable cell (filled circle, with its 95 percent confidence interval) and the change implied if the outcome had responded to the second rise as it responded to the fall after the peak (hollow diamond): the change from the peak to 2015–18, scaled by the change in the mean lagged price index at the interview dates. Each respondent's own occupation, as an indicator equal to 100 or 0, in the ten countries surveyed both in 2015–18 and in 2019–25, surveys to 2025. Stars test the estimate against the implied change, estimated as one linear combination of the coefficients so that the test includes the variance of both (Table OA2, column 3): $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Table OA2 reports the estimates. Data from the Demographic and Health Surveys (ICF, 2026). Standard errors clustered by cell.

In Figure OA1, I plot the change from 2015–18 to 2019–25 in the ten countries surveyed in both windows, against the change that the response after the peak implies (Table OA2 reports the estimates). Men's farming does not fall: it rises by 2.0 points,

which excludes both benchmarks, the fall of 7.0 points as well as that of 15.8, at the 1 percent level, and also excludes the boom benchmark with the price deflated by US consumer prices. Men's manual work does not move: the estimate is 0.1 points, and the 95 percent interval, from $-3.2$ to 3.5, rules out the rise of 10.2 points implied by the response after the peak but not the rise of 3.1 implied by the response during the boom. The estimate for manual work holds without the interviews of 2020 and 2021, when COVID-19 disrupted fieldwork, without any single country, and among men aged 15 to 24, who were children at the peak.[18] Women's sales do not rise again. Had they responded as they did at the peak, they would have risen by 5.7 points; the estimate is $-0.4$, with an interval that ends at 2.2; among all adults, manual work and sales stay flat against implied rises of 4.0 and 3.4 points. Women's current work changes by $-1.2$ points against an implied fall of 3.0.[19]

In the five countries that enter both comparisons, men's manual work changes by $-1.0$ points against an implied rise of 10.6, their farming by 3.6 against an implied fall of 15.9, and women's sales by 0.3 against an implied rise of 6.9 (Appendix Table B8, Panel D).

In Table OA4, I split the second rise by earlier mining and by age. The shortfall appears both in gold-suitable cells with an earlier record of gold or artisanal mining and in those without one: men's farming changes by 3.5 and 0.4 points, against implied falls of 11.3 and 28.4. The records are undated and exist in 44 percent of gold-suitable cells, so the split is harder to reconcile with depletion alone without ruling it out. Men aged 15 to 24 at interview, most of them children at the peak, left farming no more than older men: their farming changes by 4.8 points and that of men aged 25 and older by 0.0, against implied falls of 14.9 and 16.4.

In local currency, converted at official exchange rates and deflated by consumer prices, the response during the boom implies a fall in men's farming of 5.6 points over the second rise, which the estimate of 2.0 also excludes (Appendix Table B8, Panel E). The windows overstate the contrast, however. In Table OA3, I estimate

$$Y_{ickt} = \beta\,(S_c \times P_{t-1}) + \theta\,(S_c \times P_{t-1} \times \mathbf{1}[t \geq 2019]) + \rho\,(S_c \times \tau_t) + X'_{ickt}\gamma + \alpha_c + \lambda_{kt} + \varepsilon_{ickt}, \tag{4}$$

where $\tau_t$ is decades since 2000; Panel A sets $\theta = 0$. In the ten countries, men's farming falls by 17.2 points per unit of the price index before 2019, a response that pools the boom with the years after the peak, and by 11.4 after it (Panel B), so the response after 2019 is about a third smaller than the average response before it. Men's manual work

[18] The estimates are 0.6 points without 2020 and 2021, $-0.9$ among men aged 15 to 24, and between $-0.8$ and 1.1 when each of the ten countries is dropped in turn; the upper end of the 95 percent interval is at most 4.6 points in every case.

[19] These tests use standard errors clustered by cell. A jackknife that drops one of the ten countries at a time, with $t$ critical values on nine degrees of freedom, gives $p$-values of 0.03 for the gap between men's manual work and its implied rise of 10.2 points, 0.08 for the gap between their farming and its implied fall of 15.8, and 0.34 for women's sales.

changes less, from 9.3 points per unit of the index before 2019 to 8.0 after it, while women's sales fall from 7.2 to 4.6, by more than a third. Panel B lets the second rise change the slope on the price but not the level of gold-suitable cells relative to the trend. With a separate level term for the years from 2019, the slope change and the level shift cannot be separated in ten countries: their standard errors exceed 10 points. Measured instead as the distance from the price line in 2019–25, men's farming sits 11.6 points above it and women's sales 5.3 points below it, and with country clusters neither distance is statistically significant at the 5 percent level (Panel C).

Per unit of the dollar price, the return was faster than the departure. The price index at the interview dates rose from 0.25 in 2000–05 to 1.12 at the peak and fell only to 0.89 after it. Per unit of the index, men's farming therefore fell by 18.5 points during the boom and recovered by 40.7 after it, and their manual work rose by 6.4 points and fell by 20.8; both differences are statistically significant at the 5 percent level. A response proportional to the price would have left most of the boom's jobs in place after 2014. The contrast is larger in local currency. Converted at official exchange rates and deflated by consumer prices, the gold price fell after the peak in all eight countries but by less than the dollar price, and per unit of it men's farming fell by 22.2 points during the boom and recovered by 58.5 after it (Appendix Table B8, Panel E). Net of a linear trend for gold-suitable cells, the faster return holds only in local currency. With the yearly dollar price and the trend, men's farming after the peak sits 3.9 points above the price line, imprecisely; with the local price it sits 9.4 points above it (Table OA3, Panels A and D).

Table OA2—Work in the second rise of the gold price

| | | 2019–25 minus 2015–18 (1) | Implied by response after the peak (2) | (1) minus (2) (3) | Mean (4) | Observations (5) |
|---|---|---|---|---|---|---|
| All adults | Manual work | 0.3 | 4.0 | $-3.6^{**}$ | 12.0 | 1,072,390 |
| | | (1.0) | | (1.5) | | |
| | Agriculture | 1.4 | $-8.2$ | $9.6^{***}$ | 29.1 | 1,072,390 |
| | | (1.8) | | (3.0) | | |
| | Sales | $-0.7$ | 3.4 | $-4.0^{**}$ | 15.9 | 1,072,390 |
| | | (1.0) | | (1.6) | | |
| Men | Manual work | 0.1 | 10.2 | $-10.0^{***}$ | 21.4 | 323,711 |
| | | (1.7) | | (2.5) | | |
| | Agriculture | 2.0 | $-15.8$ | $17.8^{***}$ | 39.5 | 323,711 |
| | | (2.3) | | (4.1) | | |
| Women | Currently working | $-1.2$ | $-3.0$ | 1.8 | 53.8 | 768,233 |
| | | (1.9) | | (3.1) | | |
| | Sales | $-0.4$ | 5.7 | $-6.1^{***}$ | 18.8 | 748,678 |
| | | (1.3) | | (2.2) | | |

*Notes*: Column 1 is the difference between the 2019–25 and 2015–18 coefficients of equation (2) in a fully gold-suitable cell, in percentage points, for each respondent's own occupation, as an indicator equal to 100 or 0, in the ten countries surveyed both in 2015–18 and in 2019–25, surveys to 2025, with age controls (and sex for all adults) and without the temperature and crop controls, which end in 2018, and with cell and country-by-year effects. Column 2 scales the change from 2015–18 to the peak, estimated in the same regression and countries, by the change in the mean lagged price index at the interview dates, 1.13 at the peak, 0.88 in 2015–18 and 1.31 in 2019–25. Column 3 is the difference between columns 1 and 2, estimated as one linear combination of the coefficients, so its standard error includes the variance of the implied change. Standard errors clustered by cell in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

Table OA3—Work and the yearly gold price

| | Men | | Women | |
|---|---|---|---|---|
| | Manual work (1) | Agriculture (2) | Sales (3) | Currently working (4) |
| *Panel A: Eight countries, surveys to 2018, dollar price* | | | | |
| Suitable × lagged price index | 6.99** | −17.54*** | 10.91*** | −1.05 |
| | (3.54) | (6.66) | (3.94) | (4.03) |
| Suitable × 2015–18 | −1.50 | 3.88 | 0.64 | 5.06* |
| | (2.61) | (4.68) | (2.95) | (2.68) |
| Suitable × trend (per decade) | −2.77 | 3.49 | −5.01 | −5.38* |
| | (2.96) | (5.42) | (3.70) | (2.99) |
| Wild bootstrap $p$, countries, Suitable × 2015–18 | [0.75] | [0.50] | [0.74] | [0.03] |
| Observations | 185,152 | 185,152 | 435,356 | 455,201 |
| *Panel B: Ten countries, surveys to 2025, dollar price: change in the slope* | | | | |
| Suitable × lagged price index | 9.33*** | −17.24*** | 7.22*** | −4.27 |
| | (2.60) | (4.18) | (2.49) | (3.23) |
| Suitable × lagged price index × 2019–25 | −1.37 | 5.80*** | −2.62** | 2.30 |
| | (1.36) | (1.90) | (1.08) | (1.78) |
| Suitable × trend (per decade) | −4.08*** | 4.14* | −1.57 | −1.22 |
| | (1.34) | (2.29) | (1.29) | (1.56) |
| Observations | 323,711 | 323,711 | 748,678 | 768,233 |
| *Panel C: Ten countries, surveys to 2025, dollar price: deviations* | | | | |
| Suitable × lagged price index | 6.40* | −13.26** | 5.38* | −0.47 |
| | (3.48) | (5.71) | (2.79) | (3.62) |
| Suitable × 2015–18 | −2.66 | 3.38 | −2.32 | 4.70** |
| | (2.22) | (3.49) | (1.54) | (2.21) |
| Suitable × 2019–25 | −4.94** | 11.62*** | −5.29*** | 6.68** |
| | (2.46) | (4.12) | (1.83) | (3.23) |
| Suitable × trend (per decade) | −0.99 | 0.15 | 0.13 | −4.70** |
| | (2.58) | (4.20) | (1.76) | (2.34) |
| Wild bootstrap $p$, countries, Suitable × 2019–25 | [0.35] | [0.11] | [0.07] | [0.20] |
| Observations | 323,711 | 323,711 | 748,678 | 768,233 |
| *Panel D: Eight countries, surveys to 2018, local real price* | | | | |
| Suitable × lagged local price index | 2.46 | −9.60 | 6.74 | 1.23 |
| | (3.78) | (6.93) | (4.33) | (5.27) |
| Suitable × 2015–18 | −4.00* | 9.43** | −1.58 | 6.51** |
| | (2.39) | (4.15) | (2.64) | (2.68) |
| Suitable × trend (per decade) | 0.51 | −3.70 | −2.46 | −7.63** |
| | (2.70) | (5.06) | (3.57) | (3.26) |
| Observations | 183,268 | 183,268 | 428,733 | 448,490 |
| *Panel E: Ten countries, surveys to 2025, local real price: change in the slope* | | | | |
| Suitable × lagged local price index | 0.02 | 0.14 | 5.05** | −3.45 |
| | (2.70) | (4.70) | (2.26) | (3.34) |
| Suitable × lagged local price index × 2019–25 | −1.66 | 6.05*** | −2.55** | 2.41 |
| | (1.25) | (1.84) | (1.06) | (1.79) |
| Suitable × trend (per decade) | 0.23 | −3.68 | −0.24 | −1.91 |
| | (1.22) | (2.27) | (1.07) | (1.30) |
| Observations | 323,711 | 323,711 | 748,678 | 768,233 |

*Notes*: Each panel is one regression per column of the respondent's own occupation or current work, in percentage points, on the suitable share times the lagged gold price index in each year (2013 equal to one), with cell and country-by-year effects. Panels A and D: the eight countries surveyed both at the peak and after it, surveys to 2018, with temperature, the crop-price index, age and age squared. Panels B, C and E: the ten countries surveyed both in 2015–18 and in 2019–25, surveys to 2025, with age and age squared. Suitable × 2015–18 and Suitable × 2019–25 are the suitable share times an indicator for those years: they measure how far the outcome sits from the line that the price predicts in those years, and they are zero if the outcome follows the price in proportion. Suitable × lagged price × 2019–25 is the change in the slope from 2019. The trend is the suitable share times decades since 2000. The local real price converts the dollar price at the official exchange rate and deflates it by the consumer price index (World Bank), indexed to 2013 within each country. Bootstrap $p$-values, in brackets, are from the wild cluster restricted bootstrap with country clusters and Rademacher weights, over all 256 draws in Panel A and all 1,024 in Panel C. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA4—The second rise by earlier mining and by age

| | Men | | Women |
|---|---|---|---|
| | Agriculture | Manual work | Sales |
| | (1) | (2) | (3) |
| *Panel A: Gold-suitable cells with an earlier record of gold or artisanal mining* | | | |
| 2019–25 minus 2015–18 | 3.5 | −0.9 | 0.1 |
| | (2.8) | (2.0) | (1.6) |
| Implied by the response after the peak | −11.3 | 8.7 | 7.0 |
| Difference | 14.8*** | −9.6*** | −7.0*** |
| | (4.7) | (3.3) | (2.6) |
| Mean of outcome | 38.3 | 22.0 | 19.3 |
| Observations | 257,748 | 257,748 | 610,339 |
| *Panel B: Gold-suitable cells without such a record* | | | |
| 2019–25 minus 2015–18 | 0.4 | 3.1 | −2.0 |
| | (3.1) | (2.5) | (1.8) |
| Implied by the response after the peak | −28.4 | 17.6 | 2.8 |
| Difference | 28.8*** | −14.6*** | −4.8 |
| | (5.9) | (3.1) | (3.6) |
| Mean of outcome | 39.1 | 21.1 | 20.2 |
| Observations | 218,462 | 218,462 | 514,697 |
| *Panel C: Men aged 15–24* | | | |
| 2019–25 minus 2015–18 | 4.8* | −0.9 | |
| | (2.8) | (2.3) | |
| Implied by the response after the peak | −14.9 | 7.9 | |
| Difference | 19.7*** | −8.8*** | |
| | (5.6) | (3.1) | |
| Mean of outcome | 33.7 | 19.0 | |
| Observations | 122,353 | 122,353 | |
| *Panel D: Men aged 25 and older* | | | |
| 2019–25 minus 2015–18 | 0.0 | 0.8 | |
| | (2.4) | (1.9) | |
| Implied by the response after the peak | −16.4 | 11.6 | |
| Difference | 16.4*** | −10.8*** | |
| | (4.0) | (2.8) | |
| Mean of outcome | 43.0 | 22.9 | |
| Observations | 201,291 | 201,291 | |

*Notes*: Each panel repeats Online Appendix Table OA2 in the ten countries surveyed both in 2015–18 and in 2019–25, surveys to 2025, in percentage points. Panels A and B keep the gold-suitable cells with, or without, an independent record of gold or artisanal mining in the data of Girard et al. (2025) (exploration records, deposits, administrative and survey records of artisanal sites, mercury emissions and sites detected in satellite images; Figure 2, Panel C), and all cells without gold-suitable land in both. Panels C and D split men by age at interview. The implied change is the change from 2015–18 to the peak, scaled by the change in the mean lagged price index at the interview dates; it is reported without a standard error. The difference is estimated as one linear combination of the coefficients, so its standard error includes the variance of the implied change. Panels C and D report men only, so column 3 is blank. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## OA3 Exposure within 10 Kilometres of Gold-Suitable Land

In Table OA5, I re-estimate Table 2 and Appendix Table B1 with exposure measured as the share of the cell within 10 kilometres of gold-suitable land (see also Ghazarian and Khan, 2025). Across all countries, agriculture falls by 5.2 points per unit of the price index, against 4.9 in Table 2, and women's sales rise by 4.1 points, against 3.8. In the eight countries surveyed on both sides of the peak, men's manual work rises by 4.8 points at the peak and gives up 4.3 after it, men's farming falls by 12.5 points and recovers 8.6, and women's sales rise by 6.1 points and fall by 3.7.

Table OA5—Work with exposure within 10 km of gold-suitable land

| | Currently working (1) | Manual work (2) | Extraction (3) | Agriculture (4) | Sales (5) | Services (6) |
|---|---|---|---|---|---|---|
| *Panel A: All adults* | | | | | | |
| Exposure × price, all countries | −2.27* | 0.89 | −0.04 | −5.24*** | 3.06*** | −0.94** |
| | (1.16) | (0.69) | (0.30) | (1.42) | (0.86) | (0.40) |
| Exposure × peak, fall countries | −6.16*** | 2.24** | 0.77 | −11.77*** | 4.93*** | −1.45** |
| | (1.89) | (1.00) | (0.49) | (2.16) | (1.31) | (0.66) |
| After minus peak, fall countries | 0.71 | −2.10*** | −0.81** | 4.77*** | −2.87*** | 0.00 |
| | (1.38) | (0.77) | (0.38) | (1.67) | (0.97) | (0.70) |
| Mean of outcome, all countries | 61.0 | 10.3 | 0.9 | 31.1 | 14.0 | 4.2 |
| Mean of outcome, fall countries | 63.6 | 10.1 | 0.9 | 29.6 | 19.7 | 4.2 |
| Observations, all countries | 1,691,736 | 1,657,666 | 1,116,355 | 1,657,666 | 1,657,666 | 1,657,666 |
| Observations, fall countries | 637,014 | 620,509 | 408,237 | 620,509 | 620,509 | 620,509 |
| *Panel B: Men* | | | | | | |
| Exposure × price, all countries | −1.23 | 1.13 | −0.12 | −5.11*** | 0.54 | −0.39 |
| | (1.42) | (0.95) | (0.53) | (1.53) | (0.65) | (0.53) |
| Exposure × peak, fall countries | −2.49 | 4.82*** | 0.99** | −12.50*** | 1.89 | −0.44 |
| | (2.76) | (1.50) | (0.41) | (3.01) | (1.21) | (0.67) |
| After minus peak, fall countries | −2.10 | −4.26*** | −0.60 | 8.64*** | −1.08 | −1.64** |
| | (1.70) | (1.22) | (0.47) | (1.96) | (0.75) | (0.68) |
| Mean of outcome, all countries | 77.3 | 18.4 | 1.9 | 41.2 | 7.6 | 5.2 |
| Mean of outcome, fall countries | 79.0 | 19.0 | 1.3 | 40.8 | 10.0 | 4.6 |
| Observations, all countries | 474,271 | 475,389 | 344,150 | 475,389 | 475,389 | 475,389 |
| Observations, fall countries | 181,812 | 185,152 | 133,282 | 185,152 | 185,152 | 185,152 |
| *Panel C: Women* | | | | | | |
| Exposure × price, all countries | −2.65* | 1.10 | 0.18 | −5.22*** | 4.13*** | −1.17** |
| | (1.42) | (0.77) | (0.31) | (1.64) | (1.14) | (0.48) |
| Exposure × peak, fall countries | −7.13*** | 1.17 | 0.75 | −10.86*** | 6.12*** | −1.86** |
| | (2.09) | (1.12) | (0.63) | (2.39) | (1.71) | (0.81) |
| After minus peak, fall countries | 1.57 | −0.79 | −0.94* | 2.57 | −3.75*** | 0.82 |
| | (1.64) | (0.92) | (0.51) | (1.87) | (1.35) | (0.87) |
| Mean of outcome, all countries | 54.7 | 7.0 | 0.5 | 27.0 | 16.5 | 3.8 |
| Mean of outcome, fall countries | 57.4 | 6.3 | 0.7 | 24.8 | 23.8 | 4.0 |
| Observations, all countries | 1,217,450 | 1,182,257 | 772,198 | 1,182,257 | 1,182,257 | 1,182,257 |
| Observations, fall countries | 455,201 | 435,356 | 274,954 | 435,356 | 435,356 | 435,356 |

*Notes*: The first row of each panel reports equation (1) and the next two rows equation (2), with exposure equal to the share of the cell within 10 km of gold-suitable land in place of the suitable share. The first row uses all countries; the other two use the eight countries surveyed both at the peak and after it. Surveys to 2018, own occupations and current work from the men's and women's interviews, indicators equal to 100 or 0. All regressions control for temperature, the crop-price index, age and age squared, with cell and country-by-year effects; Panel A also controls for sex. Currently working (column 1) is defined as in Table 2. Standard errors clustered by cell in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## OA4 Employment by Sector in the Censuses

In Table OA6, I estimate equation (1) on the IPUMS-International censuses (Minnesota Population Center, 2026), with the suitable share of each census unit in place of the cell's:

$$E_{ukt} = \beta \left( S_u \times P_{t-1} \right) + \alpha_u + \lambda_{kt} + \varepsilon_{ukt}, \tag{5}$$

where $E_{ukt}$ is the sector's share of the workers in unit $u$ of country $k$ in census year $t$ and $S_u$ the gold-suitable share of the unit, weighted by the number of workers, with standard errors clustered by unit. No country has a census both at the peak and after it, so the censuses cannot follow the cycle. Thirteen countries have at least two rounds from 1985, in 1,097 districts or regions. Men's employment in mining rises by 0.8 points per unit of the price index, on a mean share of 1.8 percent, in line with the rise of extraction in the surveys. Agriculture does not fall: its share of workers rises by 2.8 points, and that of services falls by 3.7, and with country clusters neither estimate is statistically significant. The censuses classify workers by industry, not by occupation, and over districts and regions much larger than a cell, and a digger who also farms may be counted in agriculture.

Table OA6—Artisanal gold and employment by sector in the censuses

| | Agriculture (1) | Mining (2) | Manufacturing (3) | Services (4) |
|---|---|---|---|---|
| All workers: suitable × price | 2.82* | 0.46 | −0.18 | −3.69** |
| | (1.50) | (0.29) | (0.49) | (1.44) |
| $p$, country clusters | [0.12] | [0.33] | [0.75] | [0.17] |
| Mean share (%) | 45.2 | 1.3 | 8.6 | 44.3 |
| Unit-rounds | 3,115 | 3,115 | 3,115 | 3,115 |
| Men: suitable × price | 1.32 | 0.84** | −0.14 | −3.18** |
| | (1.53) | (0.37) | (0.45) | (1.36) |
| $p$, country clusters | [0.46] | [0.28] | [0.77] | [0.27] |
| Mean share (%) | 41.5 | 1.8 | 10.3 | 45.8 |
| Unit-rounds | 3,115 | 3,115 | 3,115 | 3,115 |
| Women: suitable × price | 3.67** | 0.02 | −0.03 | −3.87** |
| | (1.69) | (0.22) | (0.67) | (1.55) |
| $p$, country clusters | [0.03] | [0.88] | [0.98] | [0.07] |
| Mean share (%) | 51.6 | 0.5 | 5.8 | 41.8 |
| Unit-rounds | 3,115 | 3,115 | 3,115 | 3,115 |
| Units | 1,097 | 1,097 | 1,097 | 1,097 |
| Countries | 13 | 13 | 13 | 13 |

*Notes*: Sample: census units (districts where the census identifies them, otherwise regions) of the IPUMS-International censuses from 1985, in the countries with at least two rounds; an observation is a unit in a census round. Outcome: the share of workers in the sector, in percent, among all workers, men or women; services include trade. Specification: equation (1), with the suitable share of the unit times the lagged gold price index (2013 equal to one), unit and country-by-year effects, weighted by the number of workers. Standard errors clustered by unit in parentheses. Brackets: $p$-values with standard errors clustered by country and $t(G-1)$ critical values; they are larger than the unit-clustered ones and are the more conservative guide. Stars use the unit-clustered standard errors: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

## OA5 Imports of Mining Machines and Mercury

In Figure OA2, I estimate, across African countries $k$ and years $t$,

$$\ln(1 + M_{kpt}) = \sum_{w \in \mathcal{W}} \beta_{wp} \left(S_k \times \mathbf{1}[t \in w]\right) + \delta_p \ln T_{kt} + \mu_{kp} + \nu_{tp} + \varepsilon_{kpt}, \tag{6}$$

where $M_{kpt}$ is the value of imports of product $p$, $S_k$ the gold-suitable share of the country, $T_{kt}$ its total imports with coefficient $\delta_p$, and $\mu_{kp}$ and $\nu_{tp}$ country and year effects; Table OA7 reports the estimates with bootstrap $p$-values. The outcome is the logarithm of one plus the value of the country's imports of each product in each year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), which reconciles the reports of importers and exporters. Exposure is the gold-suitable share of the country, the area-weighted mean of the cell shares, which ranges from 0 to 0.74 and is 0.42 in Ghana. The regressions include country and year effects and the logarithm of total imports; standard errors are clustered by country, for 54 countries. Excavator imports rise during the first rise and at the peak and do not fall afterwards (column 1); a coefficient of 2.11 at

Table OA7—Imports of mining machines and mercury by country gold exposure

| | Excavators (1) | Crushers (2) | Screens (3) | Loaders (4) | Mercury (5) | Road rollers (6) |
|---|---|---|---|---|---|---|
| 1995–99 | 0.34 | −1.11 | −0.55 | −0.84 | 0.30 | 0.51 |
| | (0.86) | (0.83) | (0.68) | (0.74) | (0.50) | (0.83) |
| Rise, 2006–11 | 1.71*** | 0.51 | −0.61 | −0.19 | 1.30** | −0.36 |
| | (0.63) | (0.46) | (0.77) | (0.63) | (0.53) | (0.81) |
| Peak, 2012–14 | 2.11** | 1.33 | 0.19 | −0.19 | 1.85** | 0.06 |
| | (1.02) | (1.10) | (1.19) | (0.96) | (0.91) | (1.05) |
| *p*, bootstrap | [0.04] | [0.23] | [0.87] | [0.87] | [0.04] | [0.95] |
| After, 2015–18 | 2.91** | 1.98* | 0.83 | 0.35 | 1.17 | 1.13 |
| | (1.16) | (1.10) | (1.20) | (0.94) | (0.87) | (1.22) |
| Second rise, 2019–24 | 3.24*** | 2.25** | 0.75 | 0.32 | 0.62 | 0.84 |
| | (1.02) | (1.08) | (1.13) | (0.89) | (0.85) | (1.02) |
| After minus peak | 0.80 | 0.65 | 0.63 | 0.53 | −0.68 | 1.07* |
| | (0.85) | (0.75) | (0.56) | (0.50) | (0.78) | (0.63) |
| *p*, bootstrap | [0.33] | [0.42] | [0.27] | [0.31] | [0.36] | [0.11] |
| Second rise minus after | 0.33 | 0.27 | −0.08 | −0.03 | −0.54 | −0.29 |
| | (0.71) | (0.63) | (0.61) | (0.45) | (0.56) | (0.75) |
| *p*, bootstrap | [0.66] | [0.67] | [0.90] | [0.95] | [0.33] | [0.70] |
| Mean of outcome | 7.48 | 6.96 | 5.96 | 7.95 | 1.17 | 6.78 |
| Country-years | 1,578 | 1,578 | 1,578 | 1,578 | 1,578 | 1,578 |
| Countries | 54 | 54 | 54 | 54 | 54 | 54 |

*Notes*: Each column regresses the logarithm of one plus the value of imports (thousand current USD) by African country and year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), on the country's gold-suitable share (the area-weighted mean of the cell share of equation (1), from 0 to 0.74) times indicators for the windows of equation (2), relative to 2000 to 2005, with country and year effects and the logarithm of total imports. HS 1992 codes: excavators with a revolving superstructure 842952; crushing and grinding machines 847420; sorting, screening, separating and washing machines 847410; front-end loaders 842951; mercury 280540; tamping machines and road rollers 842940. Standard errors clustered by country in parentheses; wild cluster restricted bootstrap *p*-values (999 Rademacher draws) in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, from country-clustered standard errors.

the peak implies that a country at Ghana's share imported 2.4 times its imports of 2000–05, relative to a country without gold-suitable land. Crushing machines follow the same path, imprecisely (column 2), while screening and washing machines and loaders do not respond (columns 3 and 4). Mercury rises during the rise and at the peak and falls back afterwards (column 5); exporter bans from 2011 and the Minamata Convention, in force from 2017, push the recorded trade down after the peak as well. Road rollers do not respond at the peak but rise after it (column 6). The estimates change little without South Africa, with quantities in place of values, or with exposure above the median as a binary indicator; without the control for total imports, road rollers rise at the peak as well.

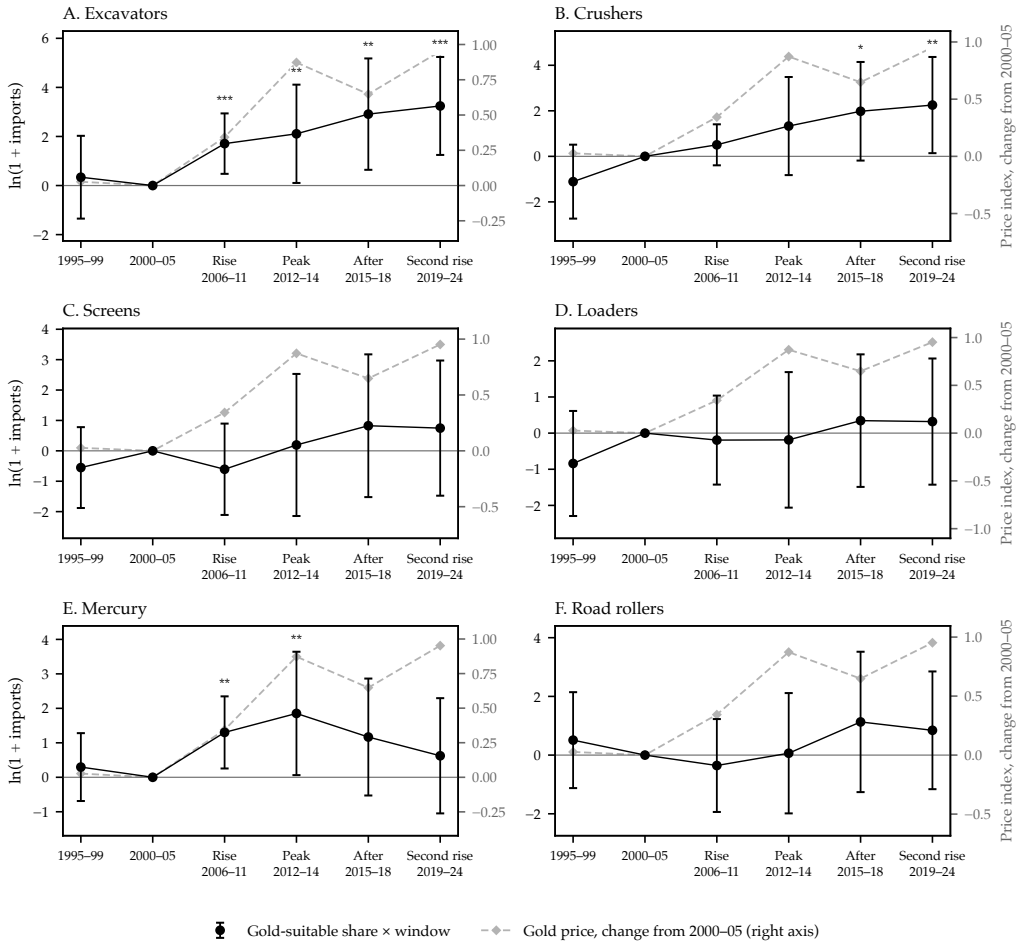



Figure OA2—Imports of mining machines and mercury by country gold exposure

*Notes*: Coefficients of equation (6) across African countries, relative to 2000–05: the logarithm of one plus the value of the country's imports of each product in each year, 1995 to 2024, from CEPII BACI (Gaulier and Zignago, 2010), on the country's gold-suitable share times indicators for each window, with country and year effects and the logarithm of total imports, for 54 countries; Table OA7 reports the estimates, bootstrap $p$-values and the HS codes. The dashed line (right axis) is the mean lagged gold price index in each window minus its mean in 2000–05 (2013 equal to one): 0.03 in 1995 to 1999, 0.34 during the rise, 0.87 at the peak, 0.65 after it and 0.95 in the second rise. Bars are 95 percent confidence intervals from standard errors clustered by country. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.